AWS CLOUD & DATA ENGINEERING

Assignment Report

Submitted by

NAME	:	Kaviya
ROLL NUMBER	:	23BTRCL049
COLLEGE NAME	:	Jain (Deemed-to-be University)
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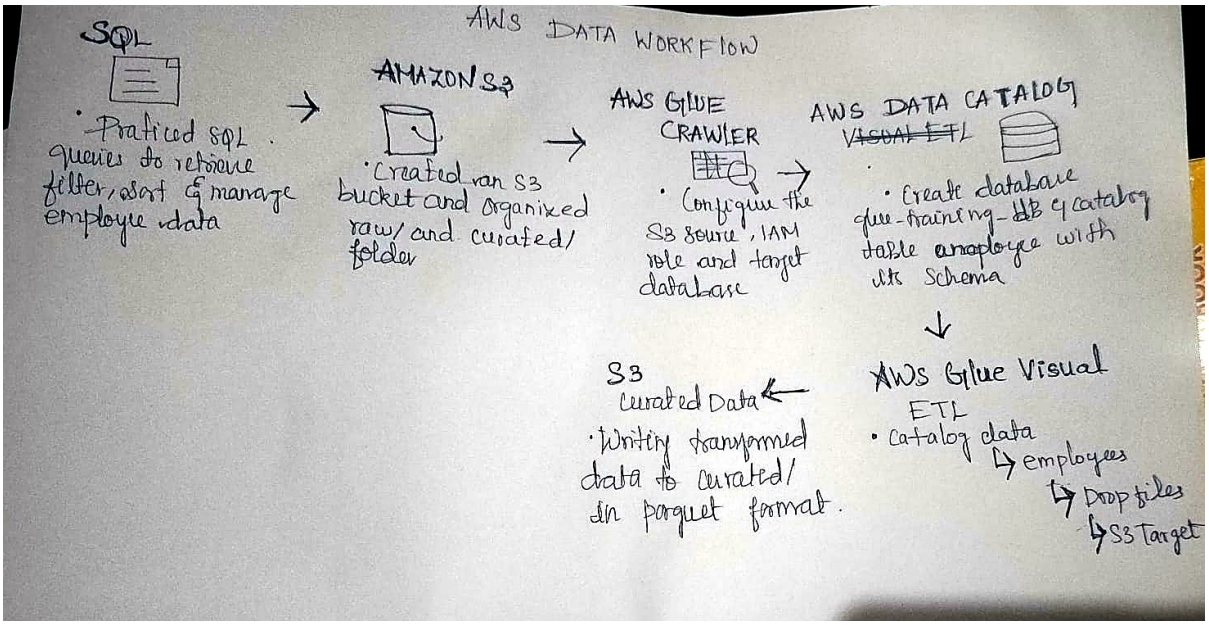


TITLE

AWS Cloud & Data Engineering

AWS Data Workflow

The diagram below summarises the complete workflow followed in this assignment, from SQL practice to the curated Parquet output in Amazon S3.

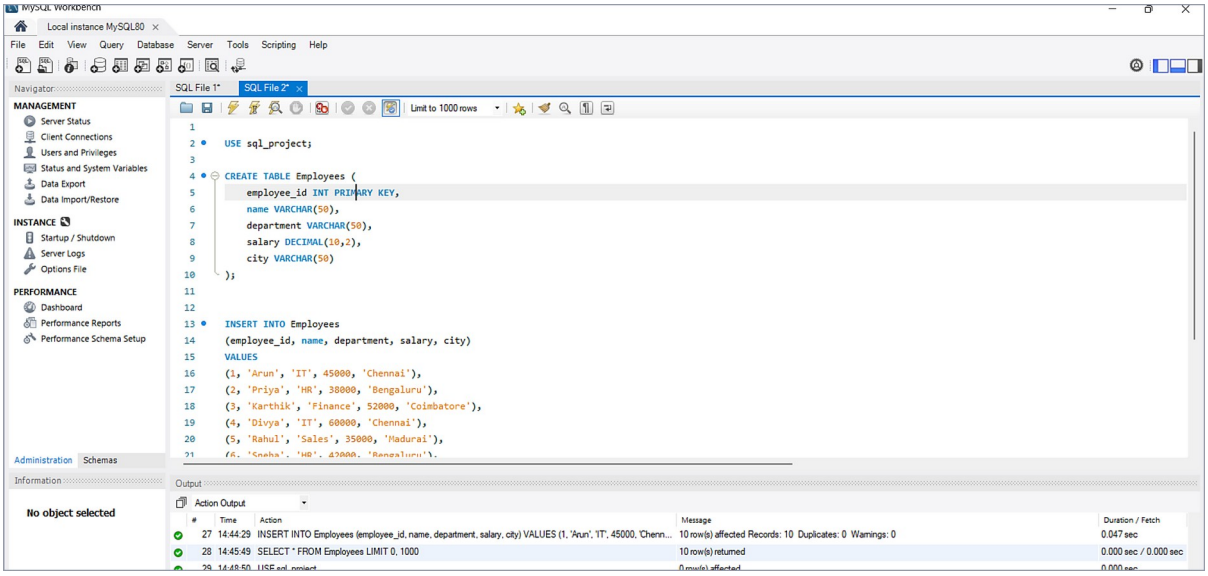


AWS Data Workflow: SQL → Amazon S3 → Glue Crawler → Data Catalog → Visual ETL → S3 Curated Data

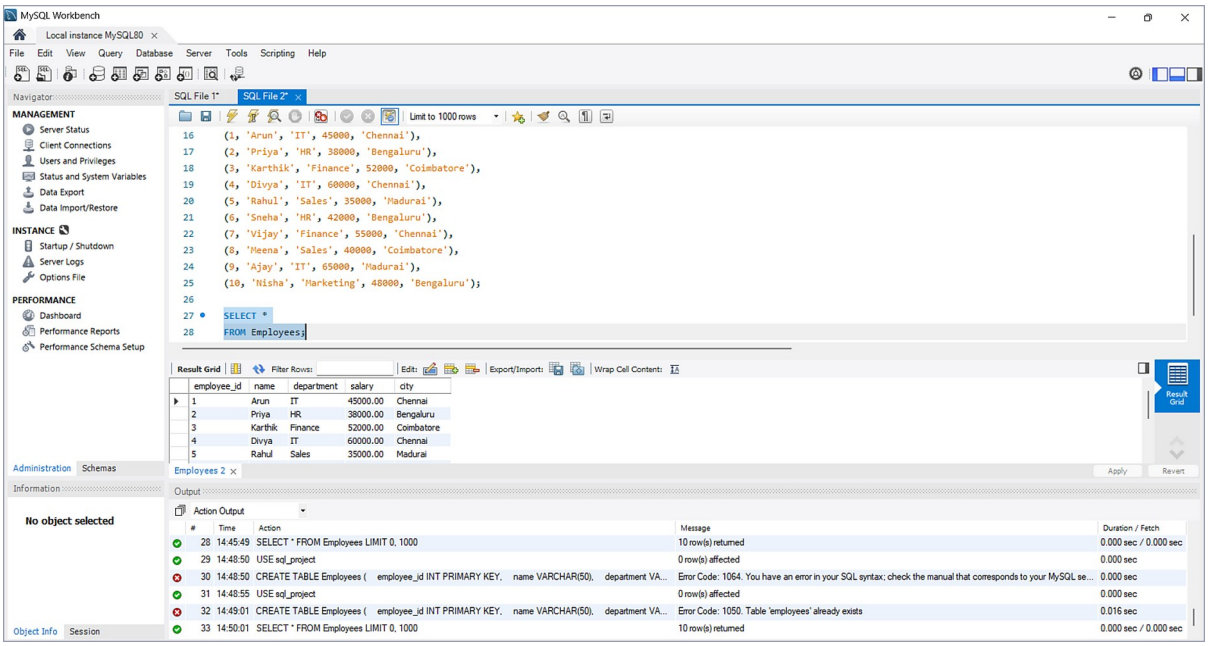
1. SQL Practice

SQL (Structured Query Language) is used to create, manage, retrieve, and manipulate data stored in relational databases. In this section, I practiced 15 SQL queries to understand basic database operations, data filtering, sorting, aggregation, joins, and subqueries. Each query and its output are shown in the MySQL Workbench screenshots below.

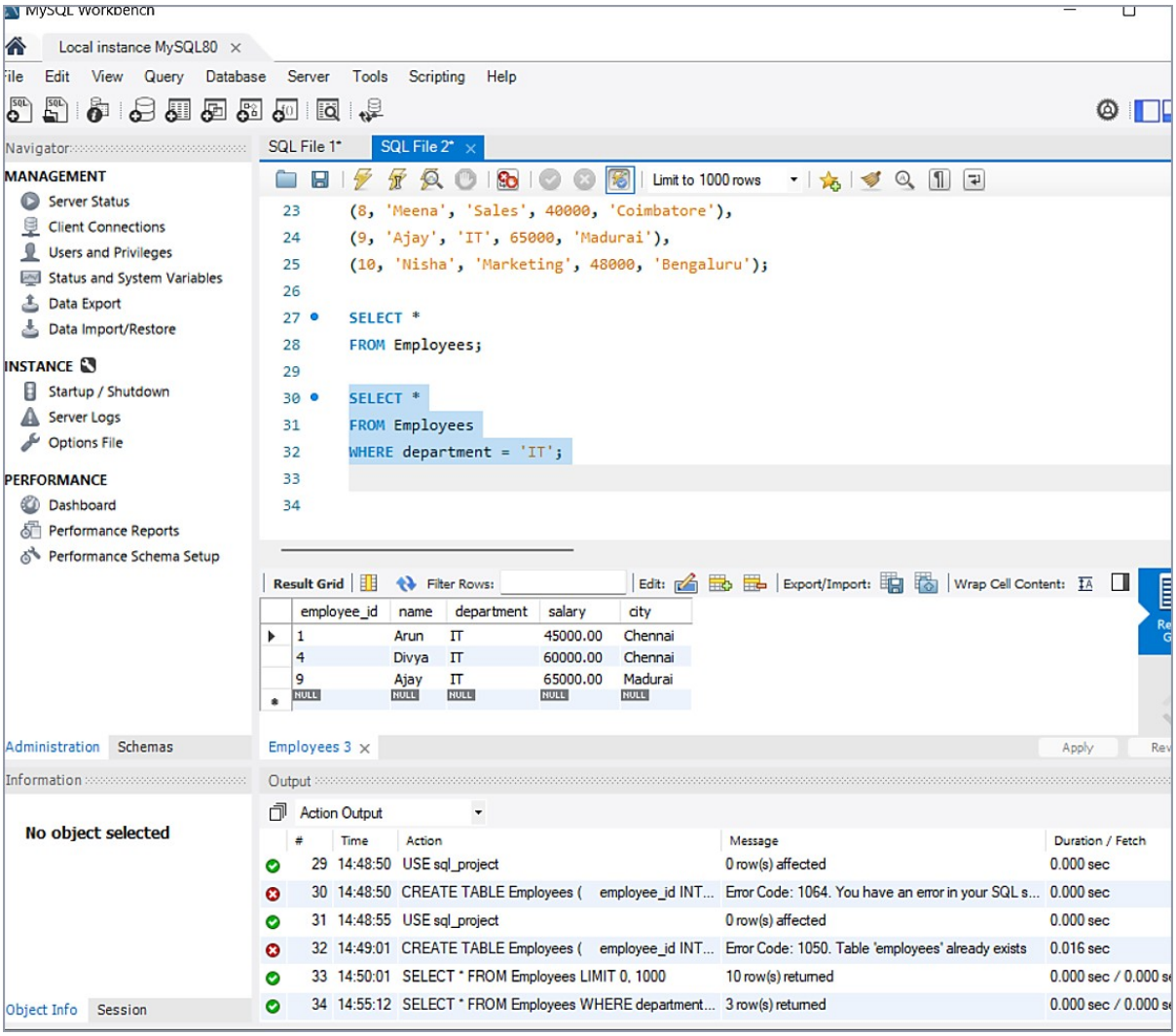
Setup: creating the Employees table and inserting records



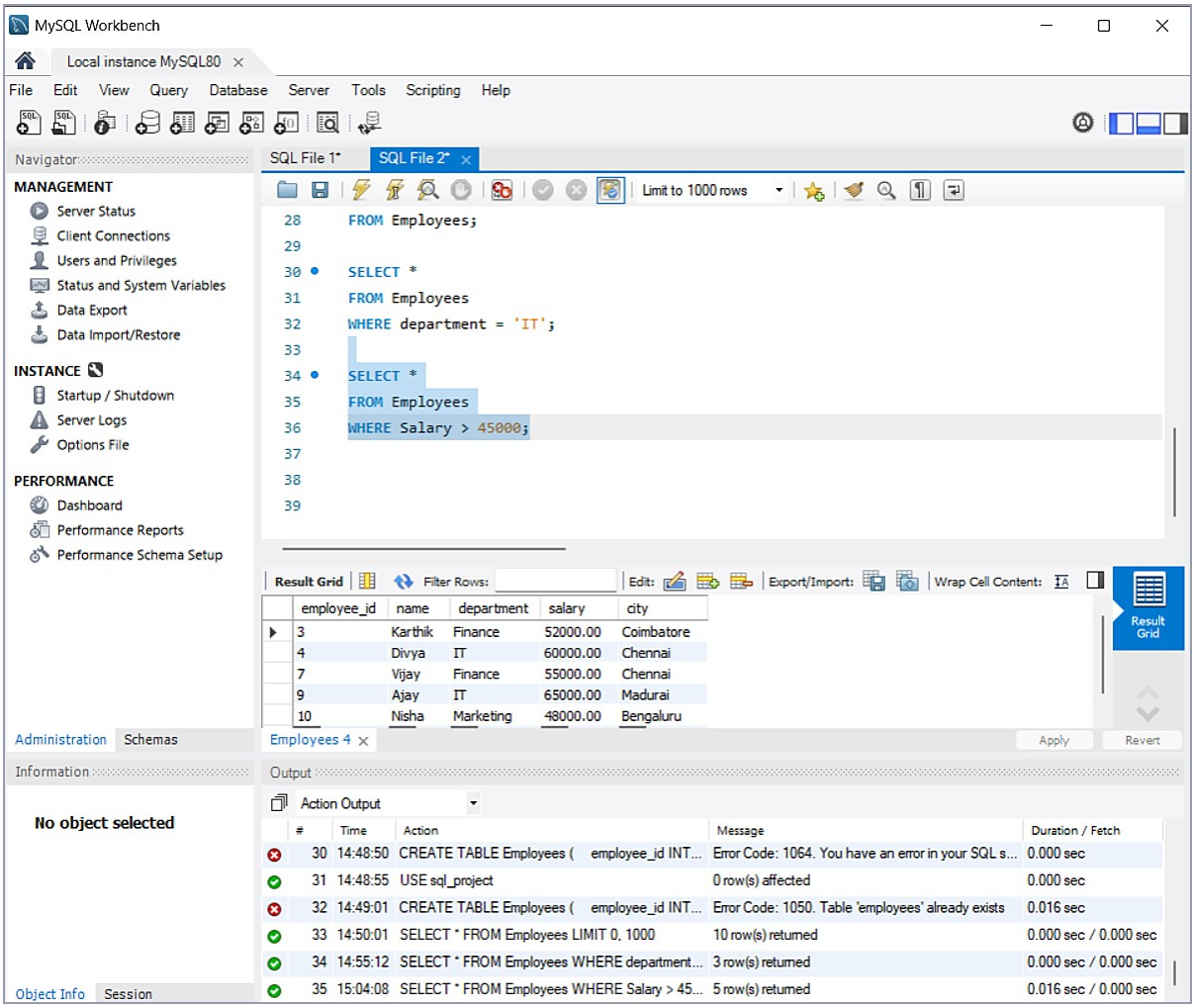
Query 1: Display all employee records



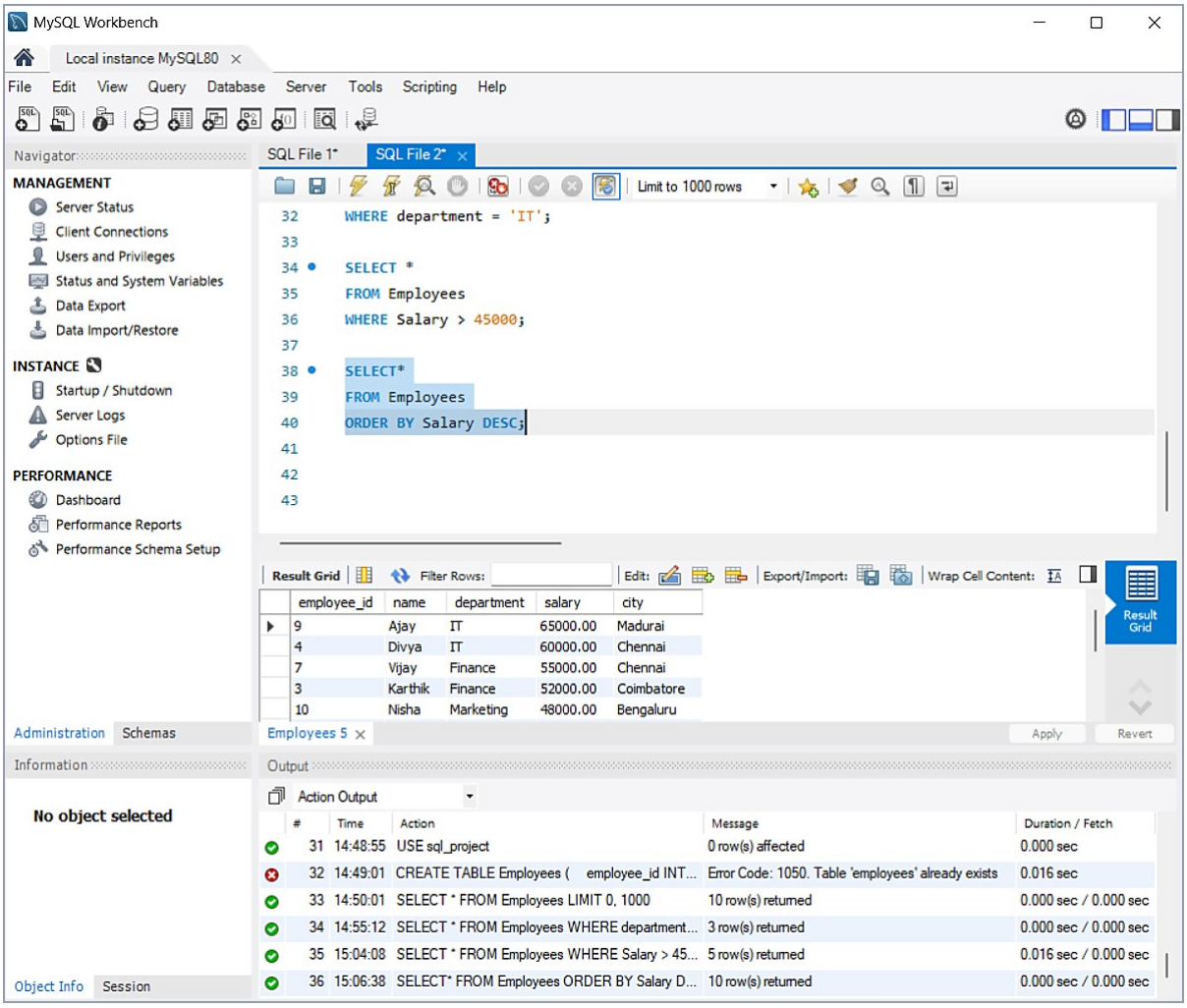
Query 2: Display employees from the IT department



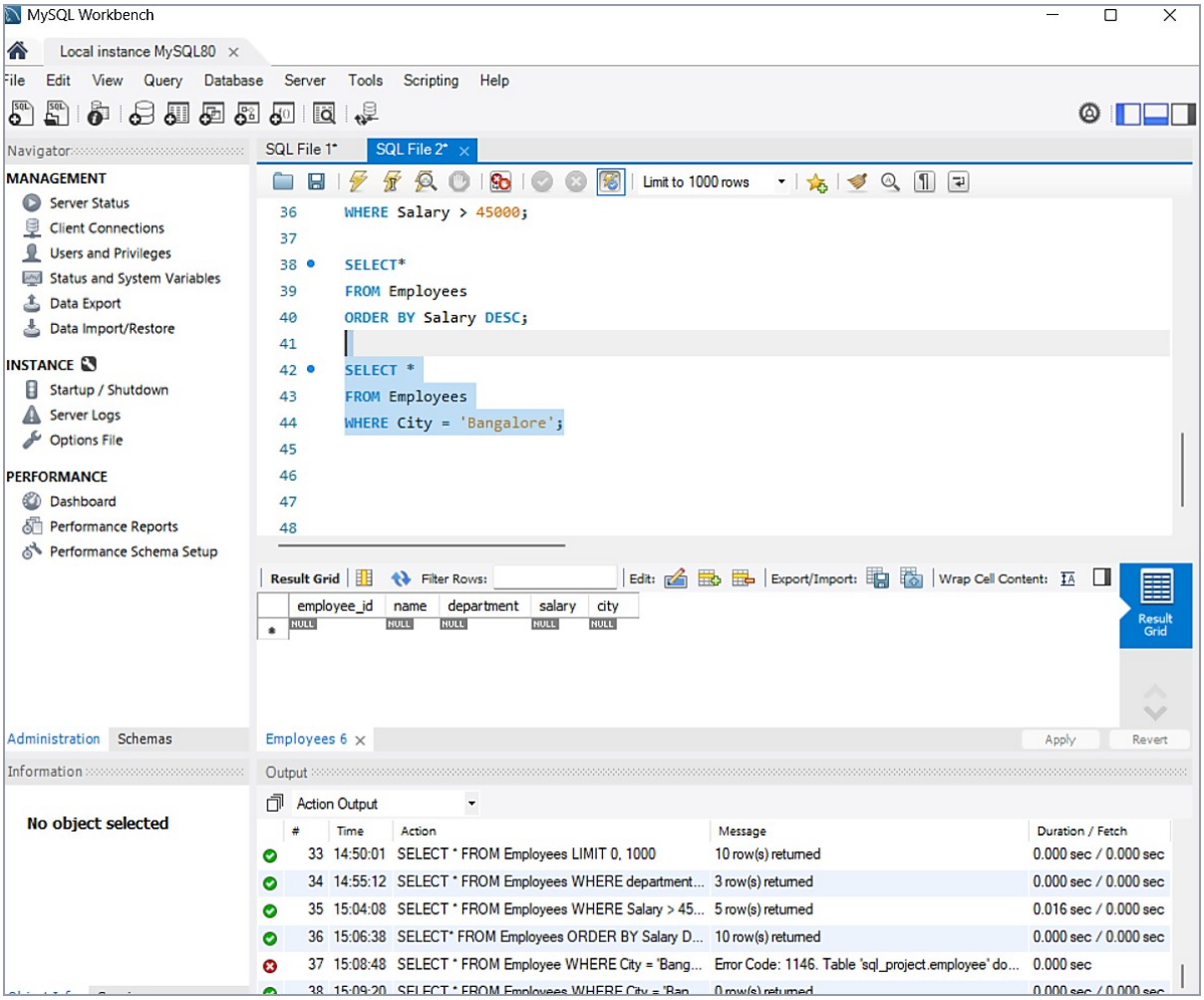
Query 3: Find employees earning more than 45,000



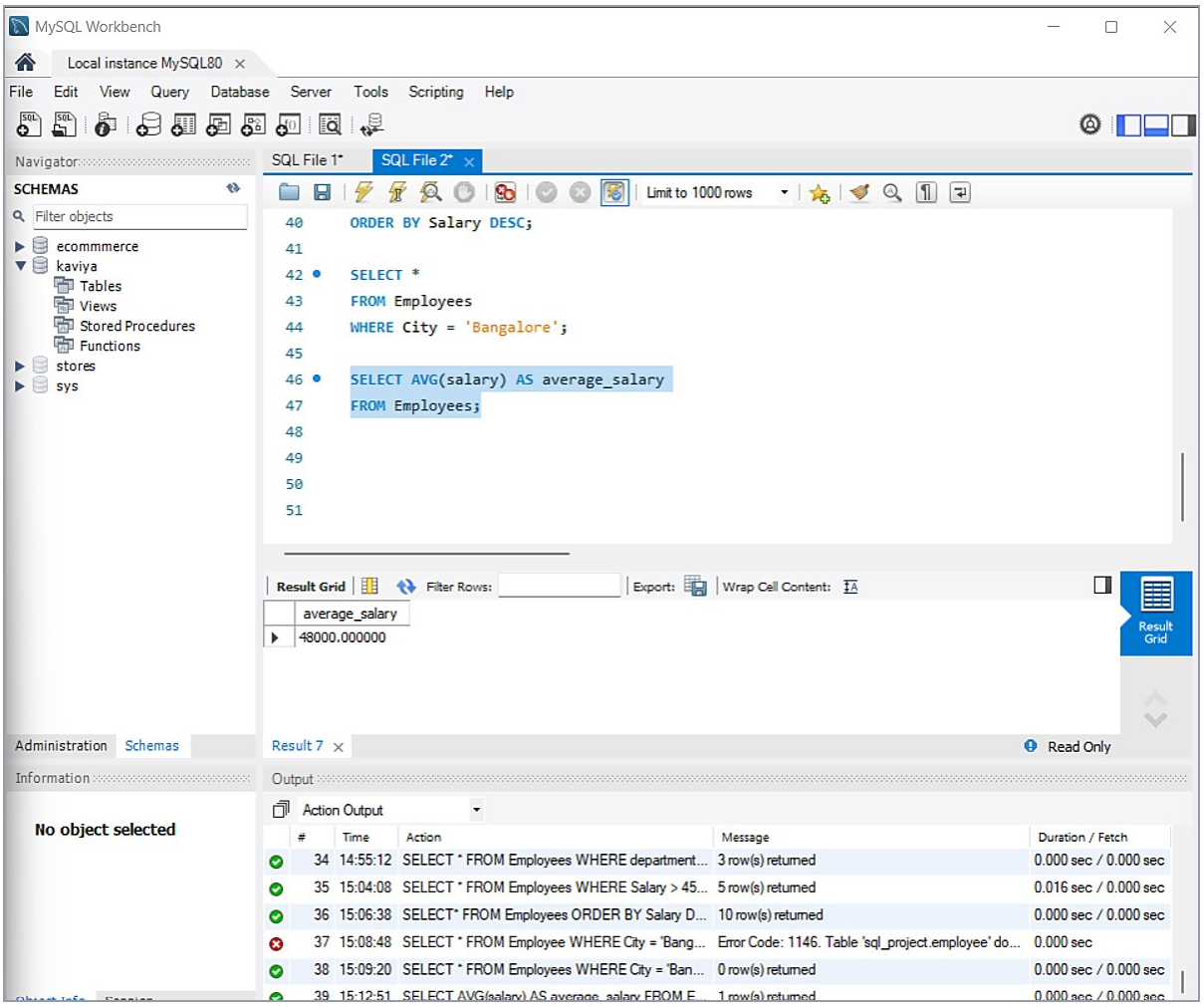
Query 4: Display all employees in descending order of salary



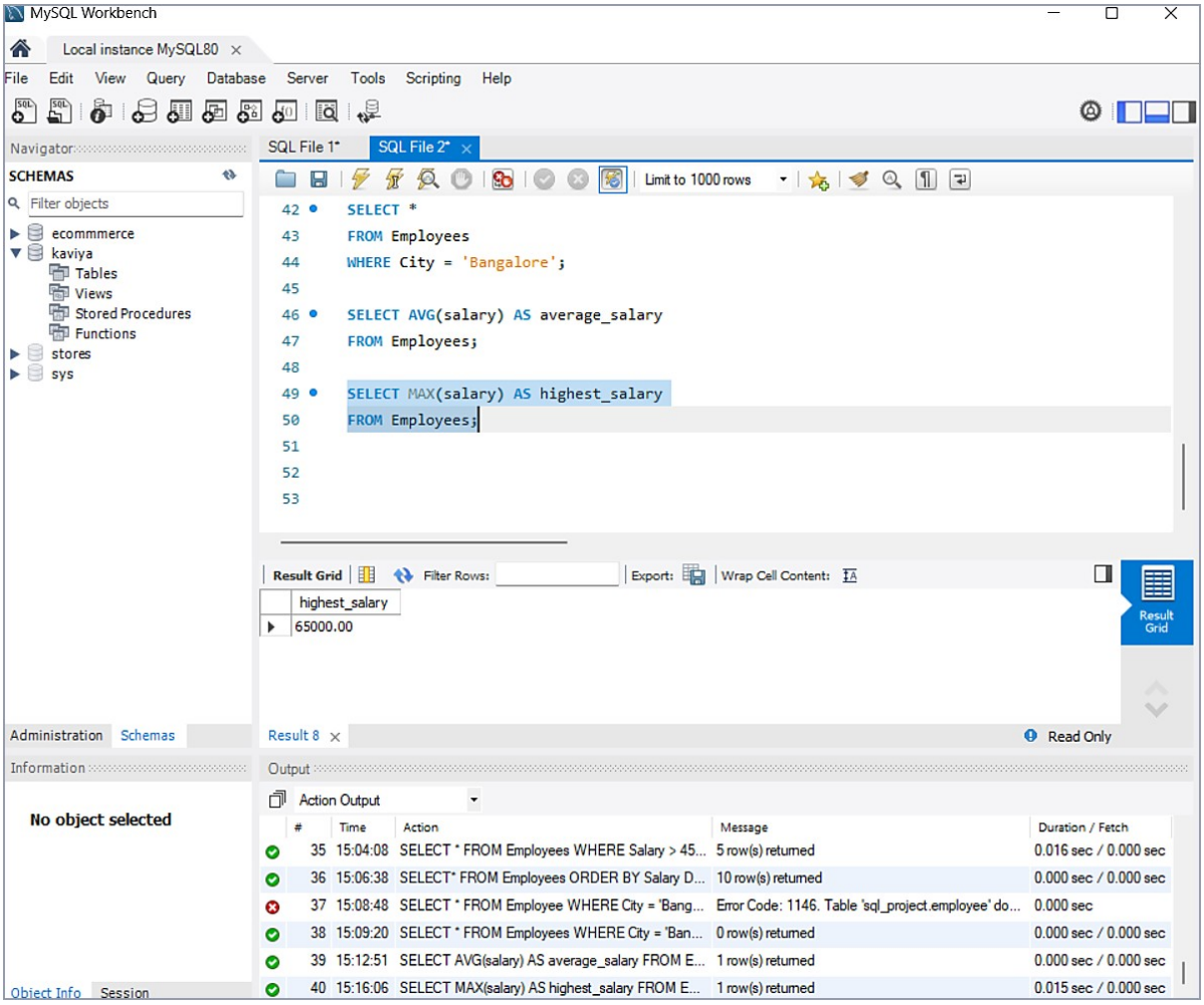
Query 5: Display employees who work in Bengaluru



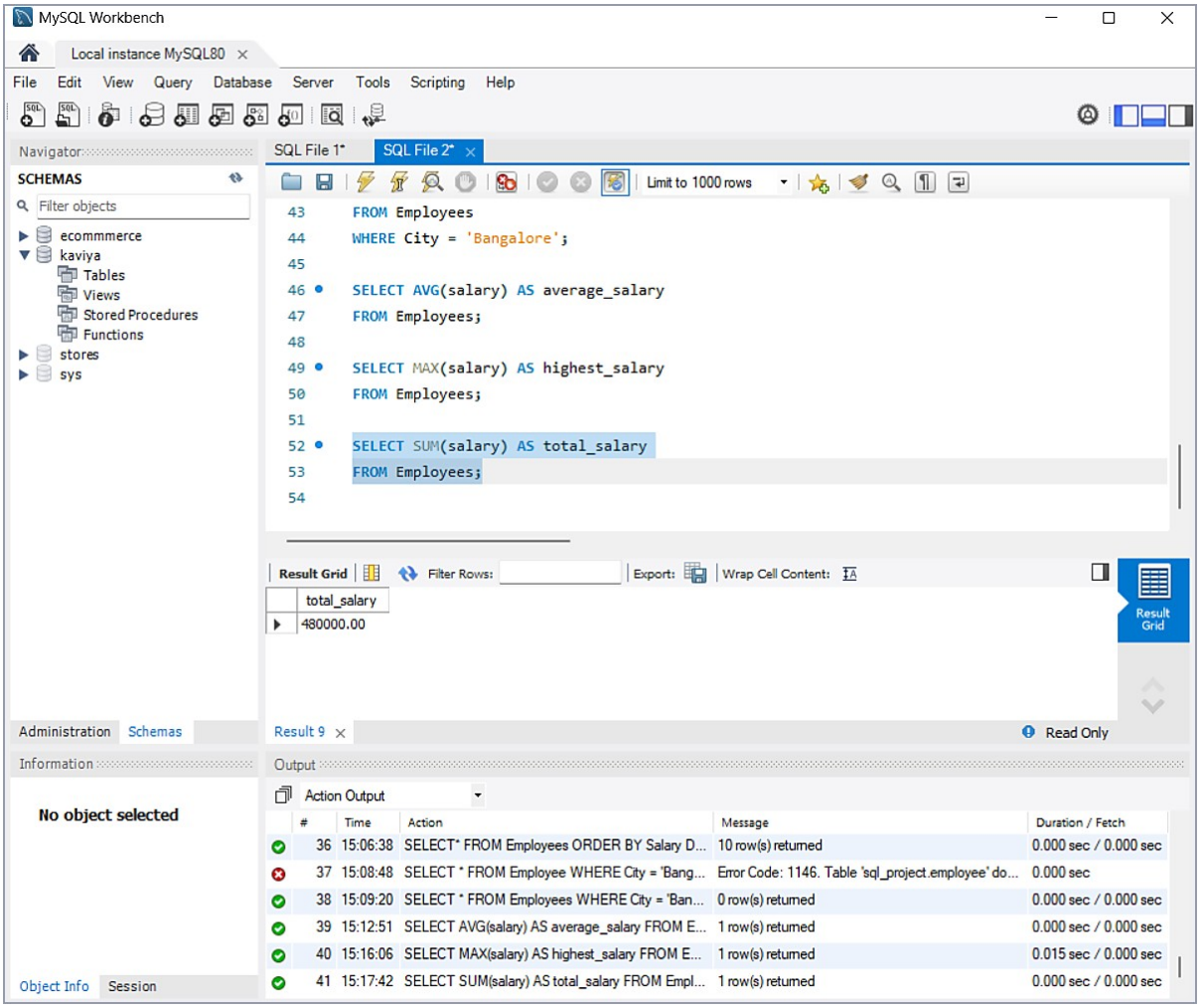
Query 6: Calculate the average salary of all employees



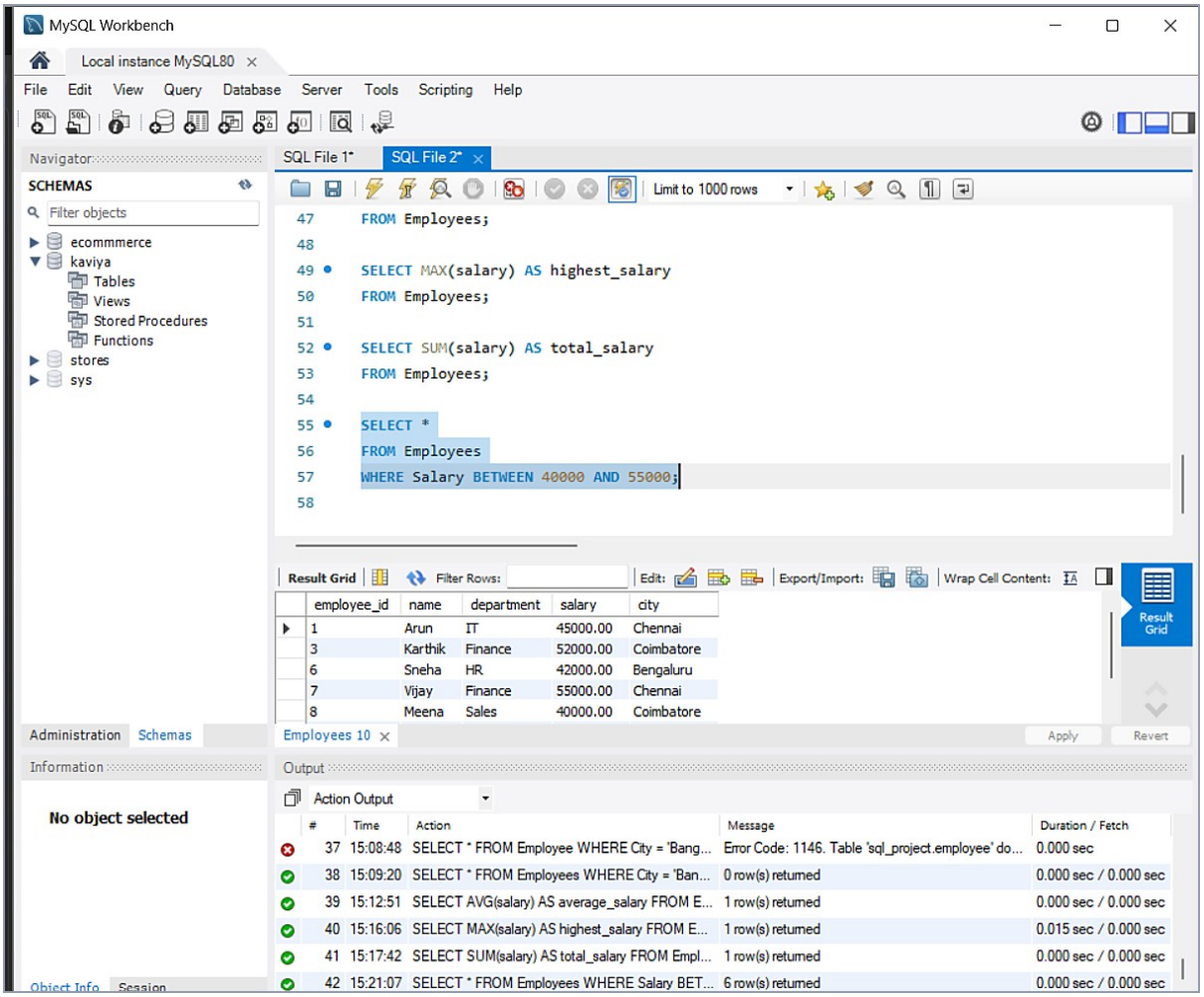
Query 7: Find the highest salary among all employees



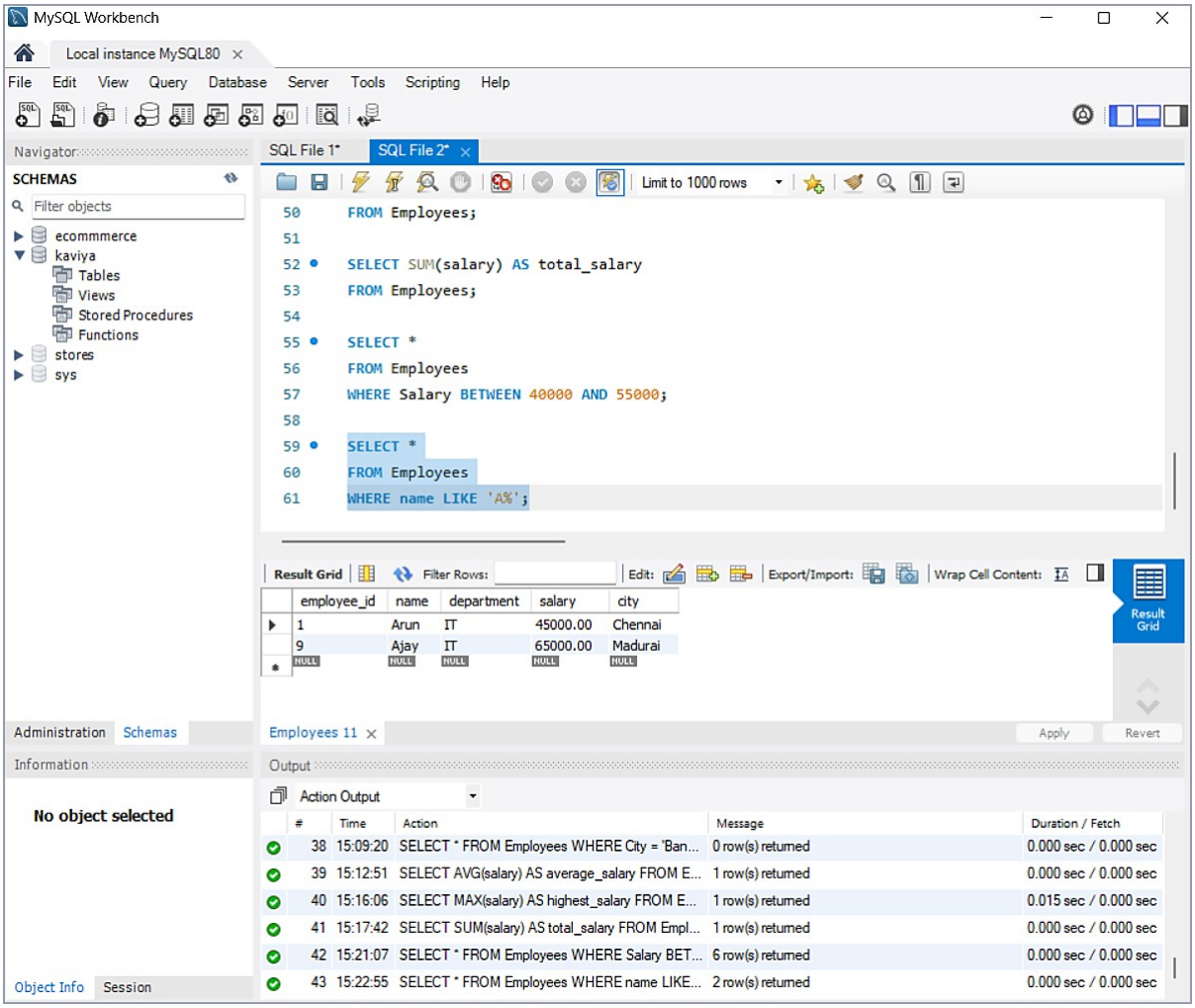
Query 8: Calculate the total salary of all employees



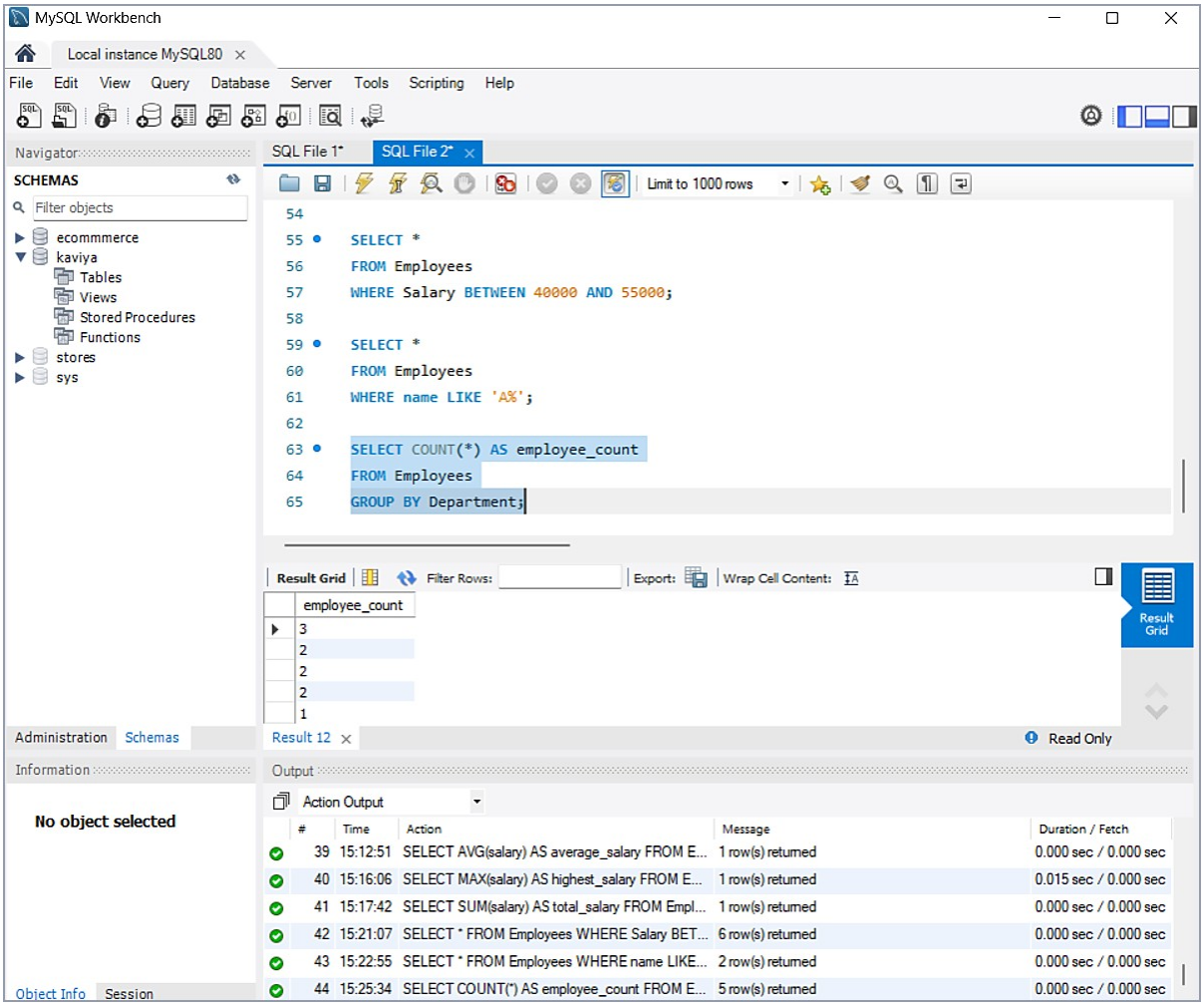
Query 9: Display employees whose salary is between 40,000 and 55,000



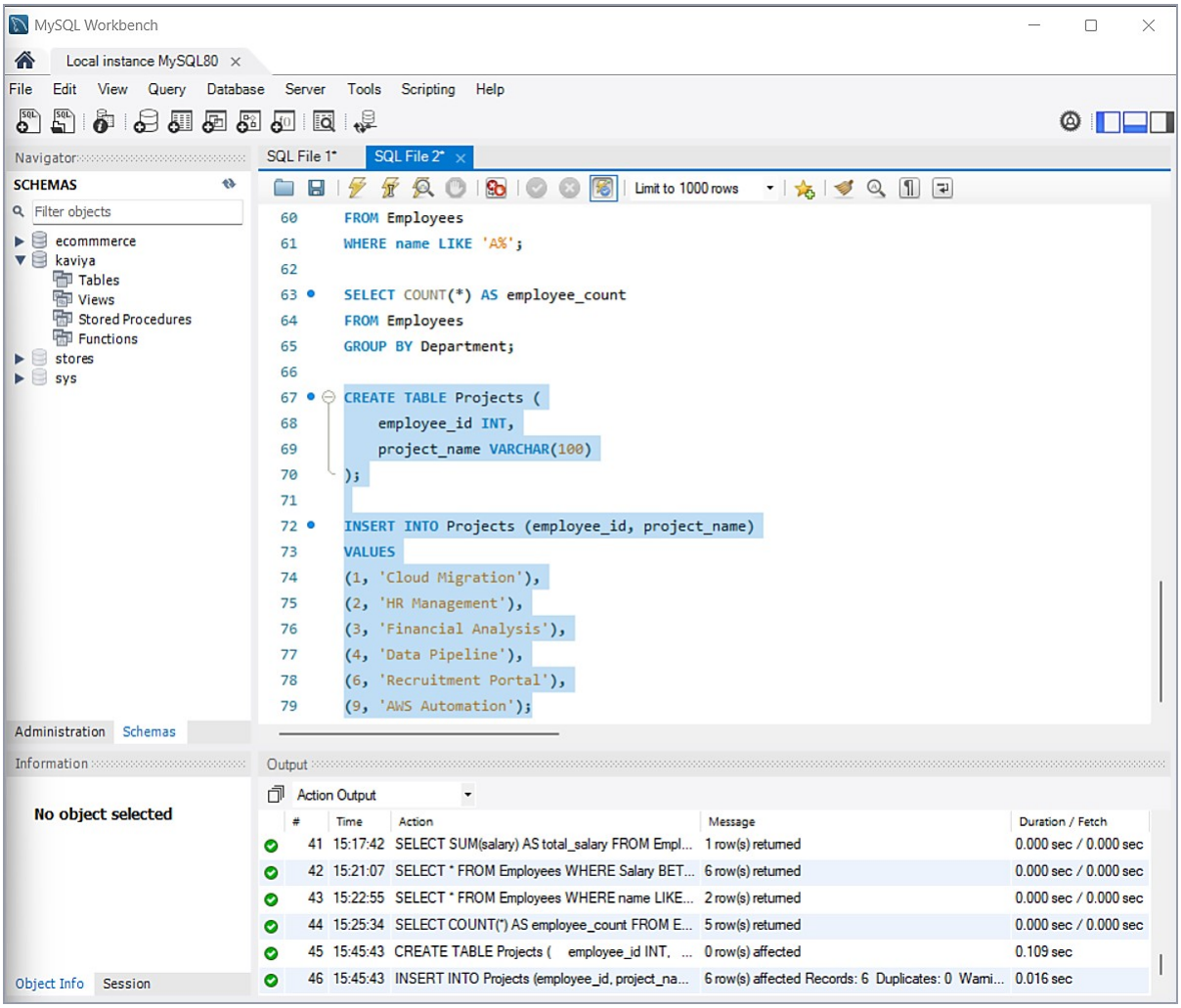
Query 10: Display employees whose names start with the letter 'A'



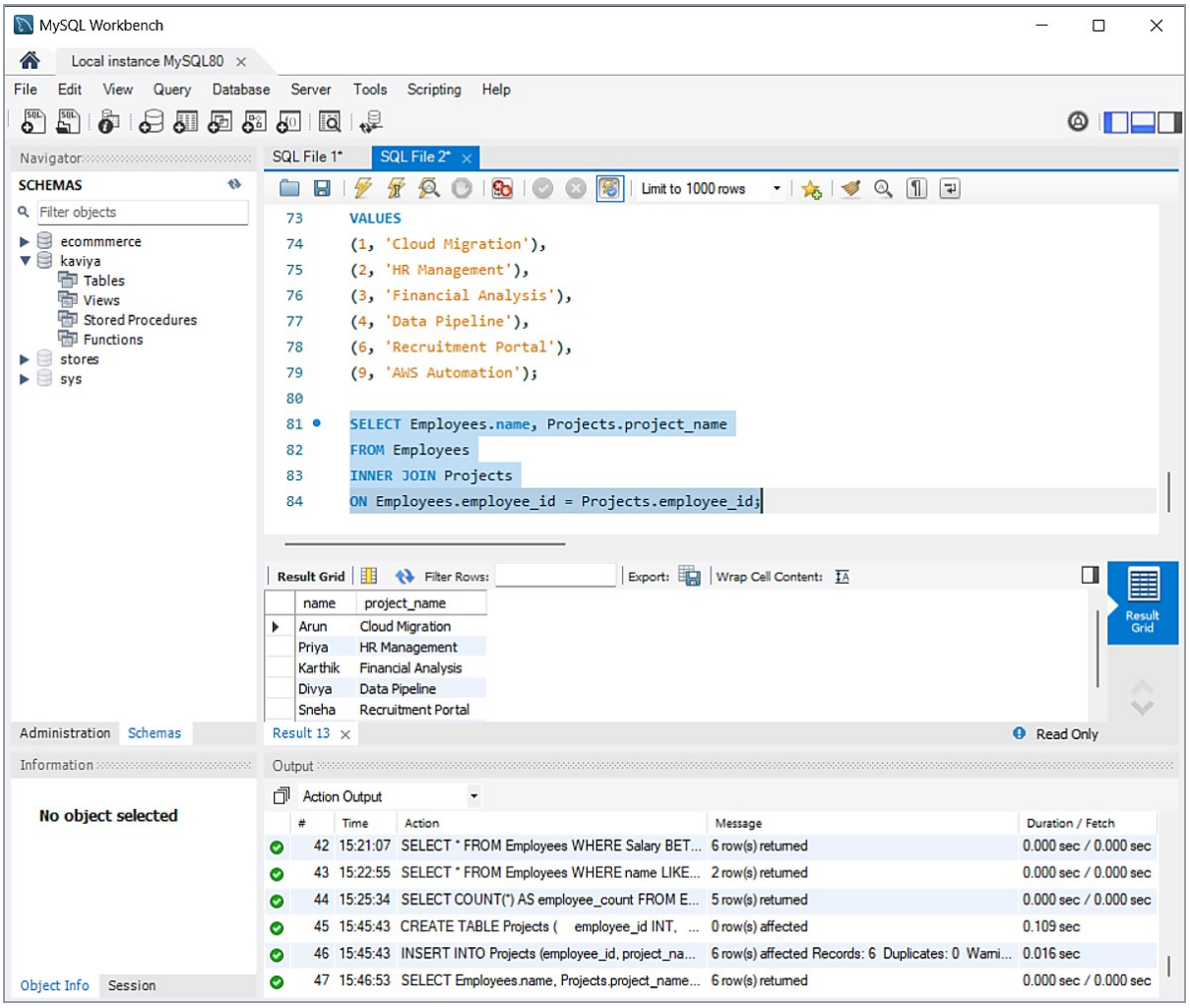
Query 11: Count the number of employees in each department



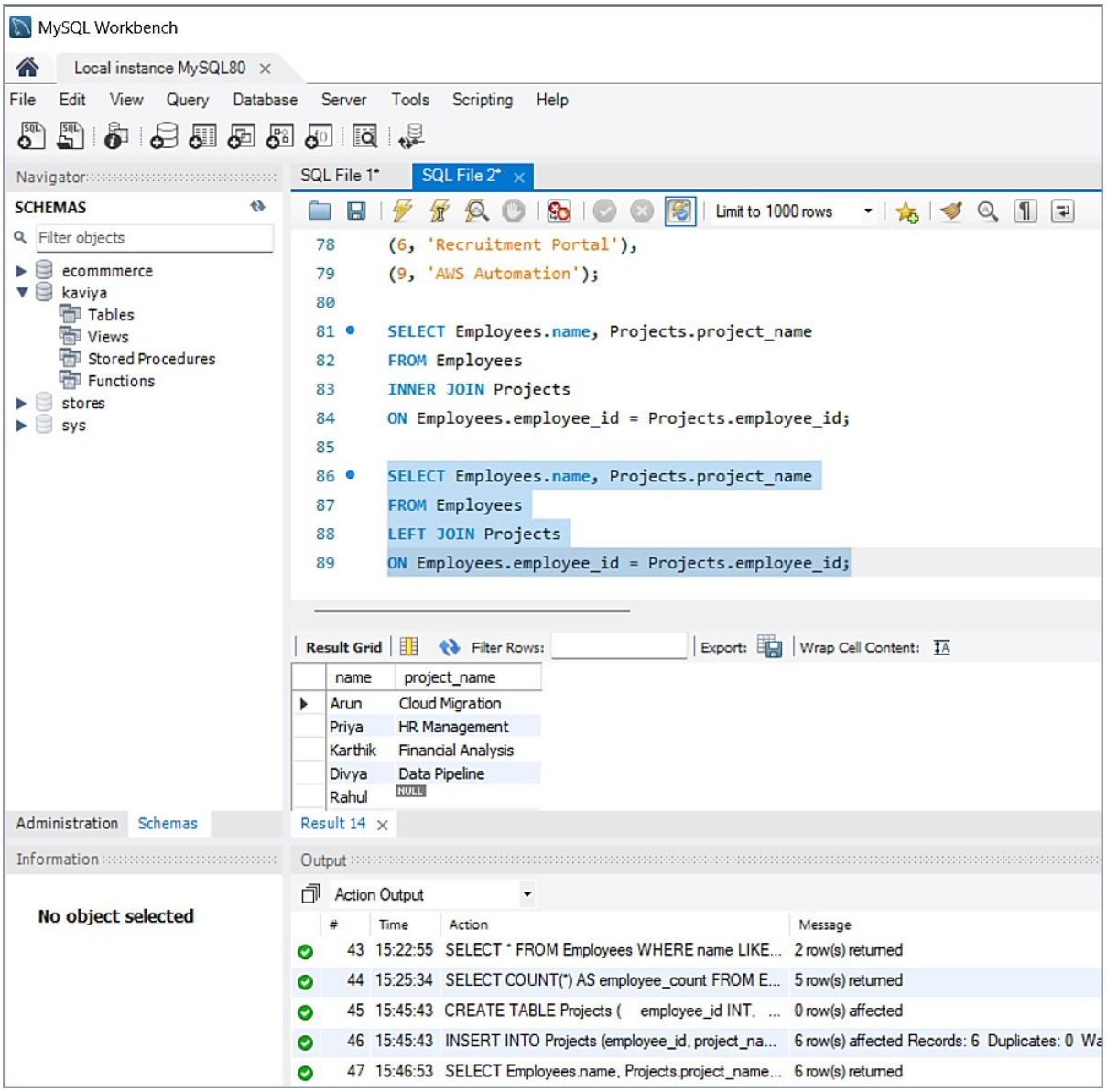
Setup: creating the Projects table (used for the join queries)



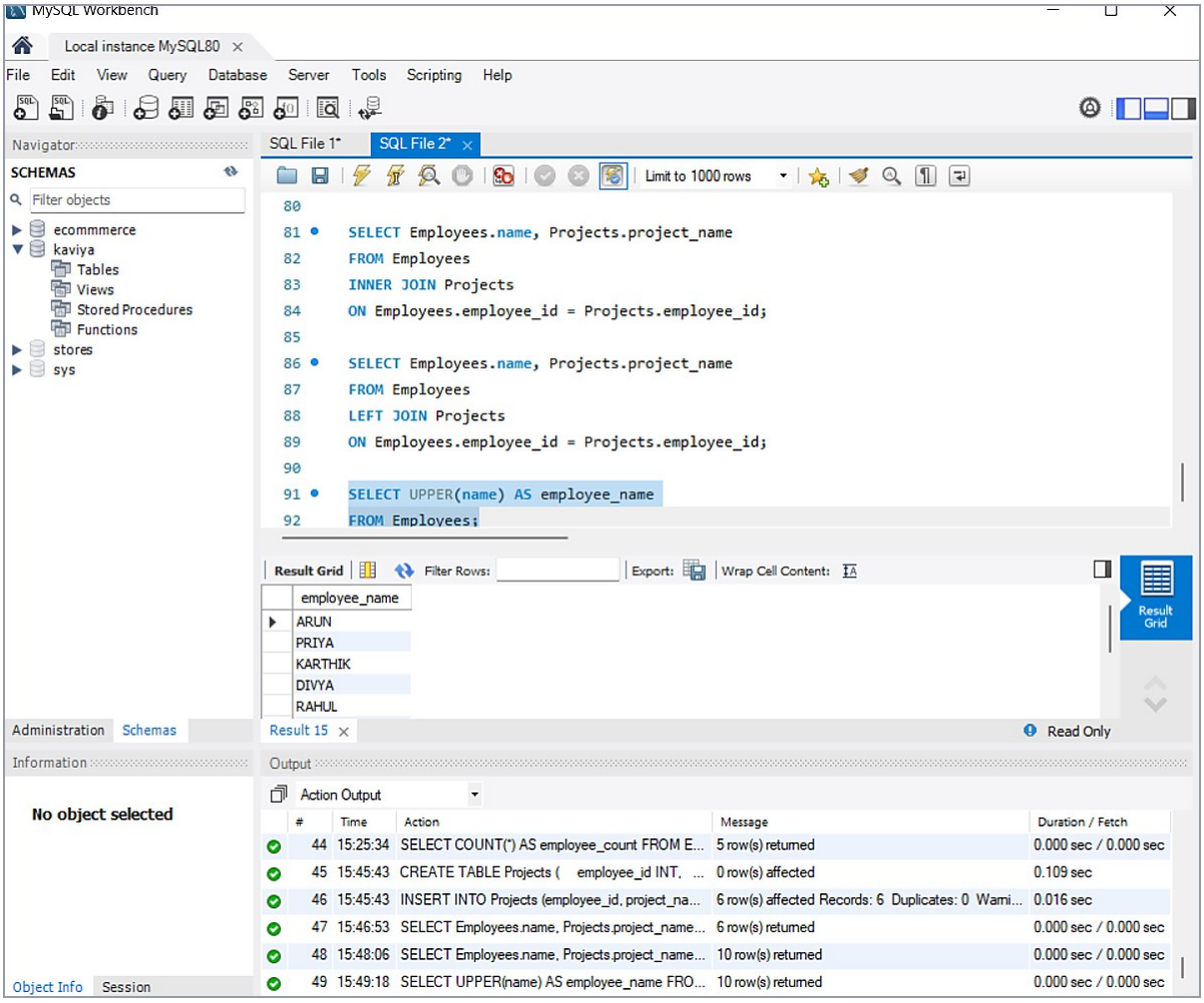
Query 12: Display employee names with their project names using INNER JOIN



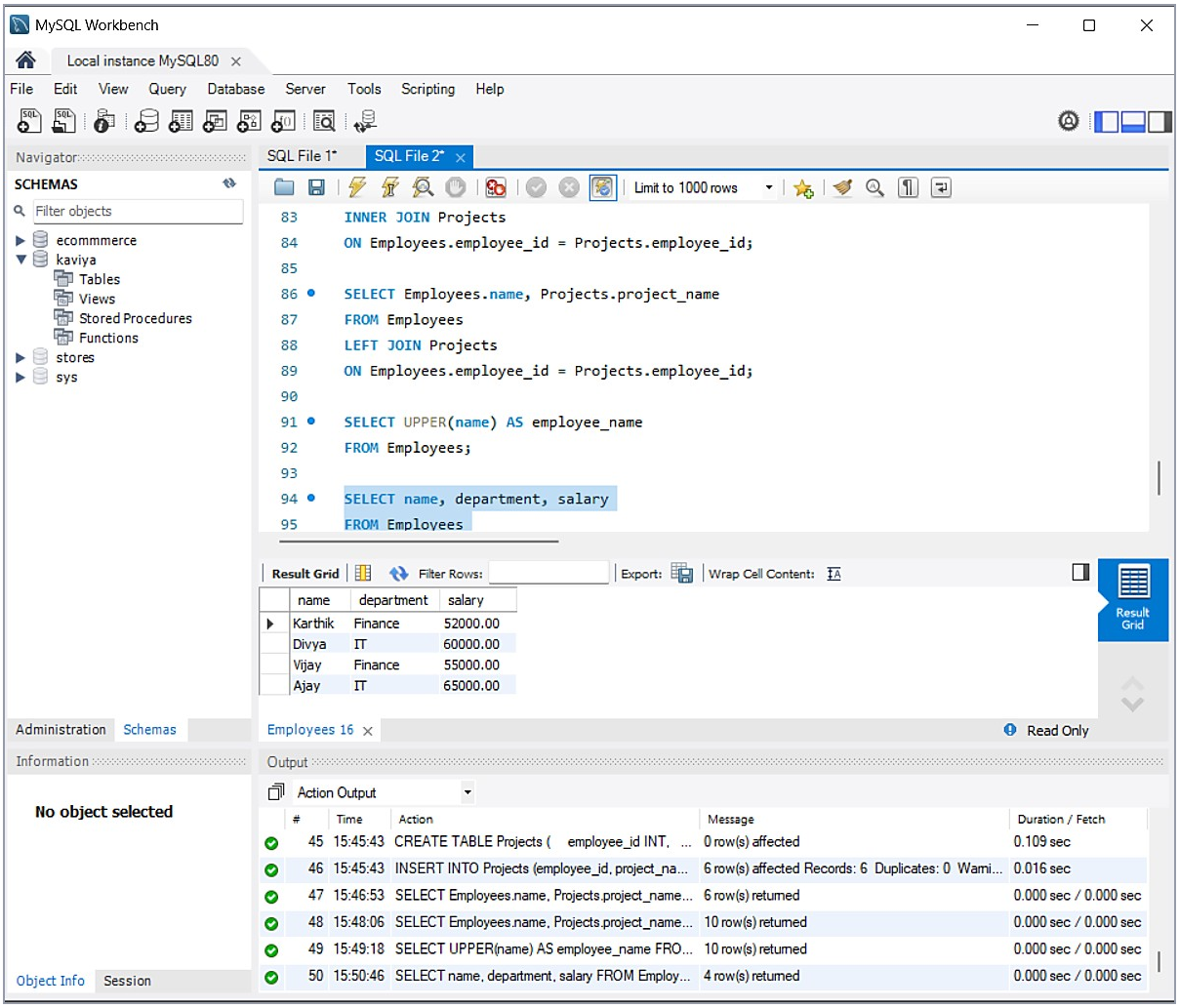
Query 13: Display employee names and project names using LEFT JOIN



Query 14: Display employee names in uppercase



Query 15: Find employees whose salary is greater than the average salary (subquery)

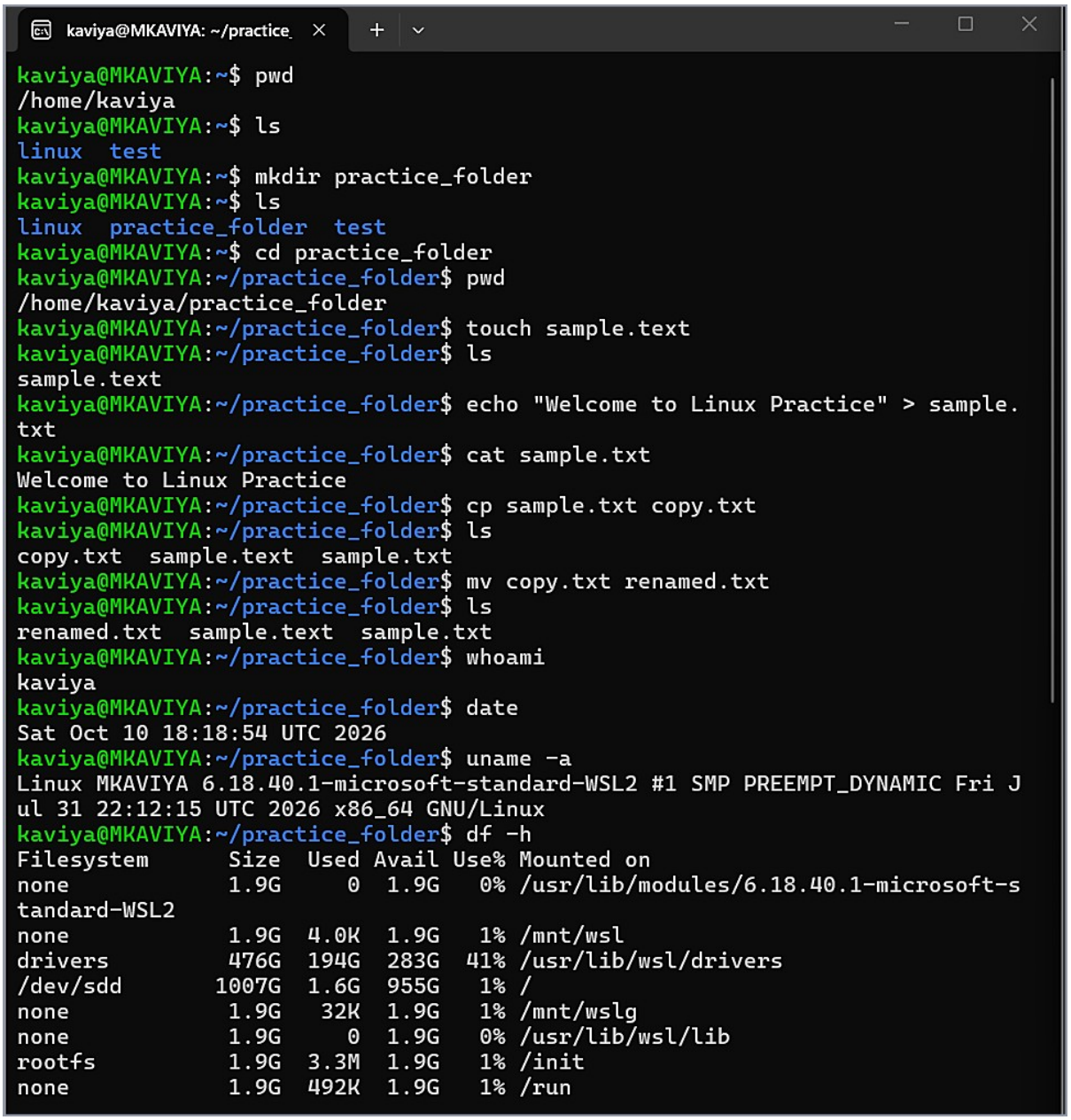


2. Linux Practice

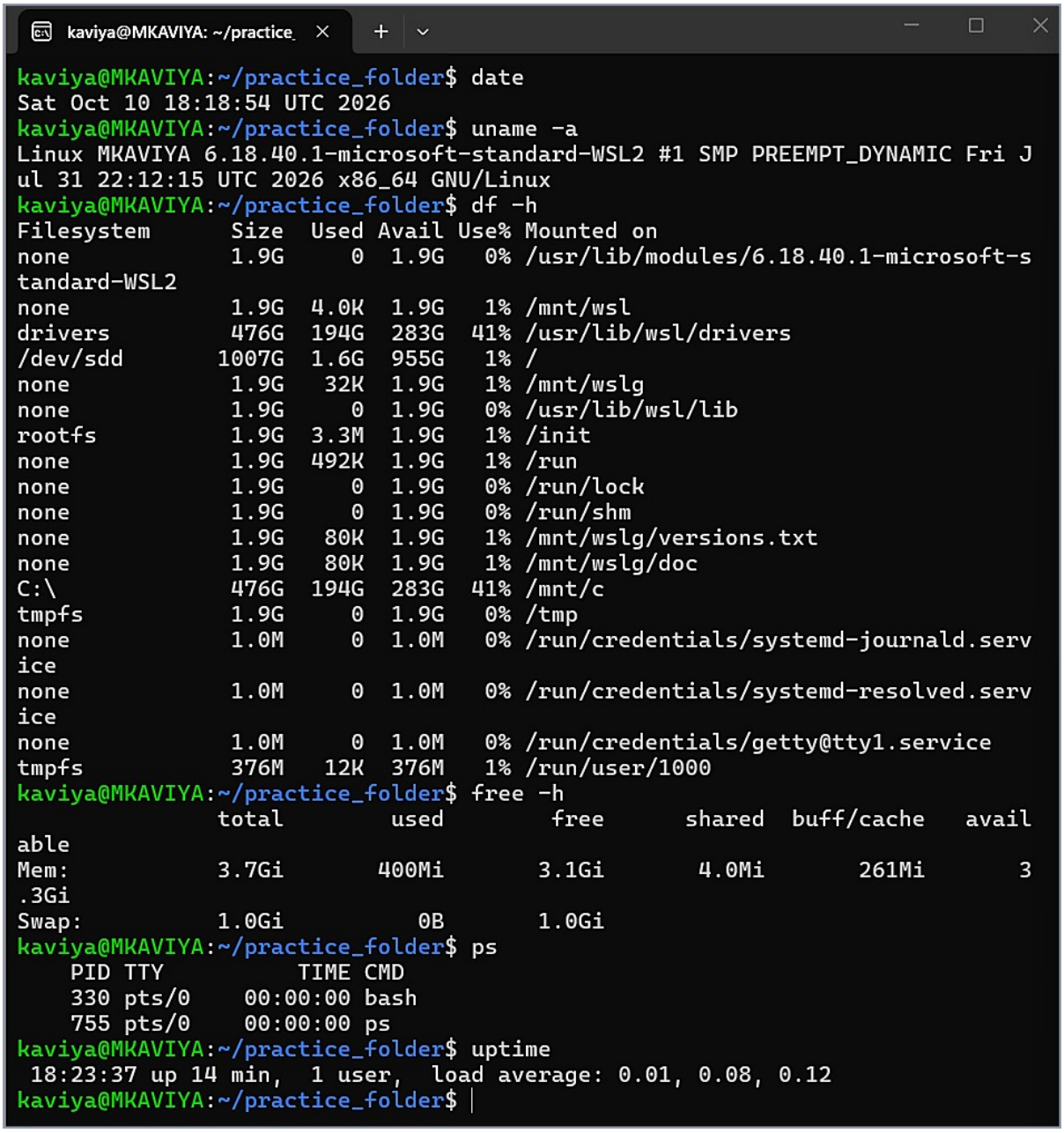
The Linux commands were run in a WSL (Ubuntu) terminal. The commands performed are summarised below, followed by the terminal screenshots showing each command and its output.

Command	Purpose
`pwd`	Print the present working directory
`ls`	List files and directories
`mkdir practice_folder`	Create a new directory
`cd practice_folder`	Change into the directory
`touch sample.text`	Create an empty file
`echo "..." > sample.text`	Write text into a file
`cat sample.text`	Display the contents of a file
`cp sample.txt copy.txt`	Copy a file
`mv copy.txt renamed.txt`	Rename (move) a file
`whoami`	Display the current username
`date`	Display the current date and time
`uname -a`	Display system information
`df -h`	Display disk usage in human-readable form
`free -h`	Display memory usage
`ps`	Display running processes
`uptime`	Show system running time and load averages

Terminal output (part 1): pwd, ls, mkdir, cd, touch, echo, cat, cp, mv, whoami, date, uname, df



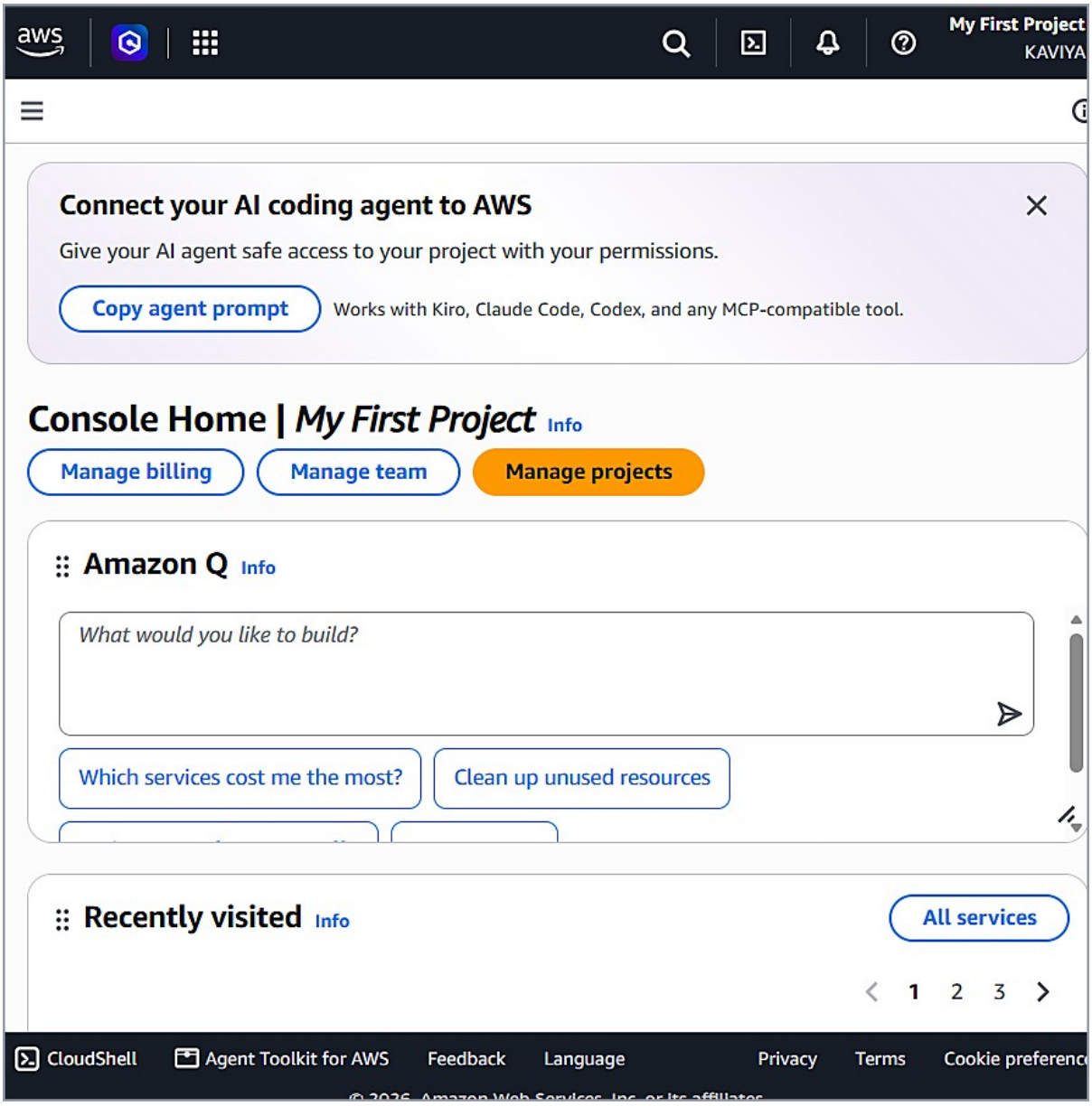
Terminal output (part 2): date, uname -a, df -h, free -h, ps, uptime



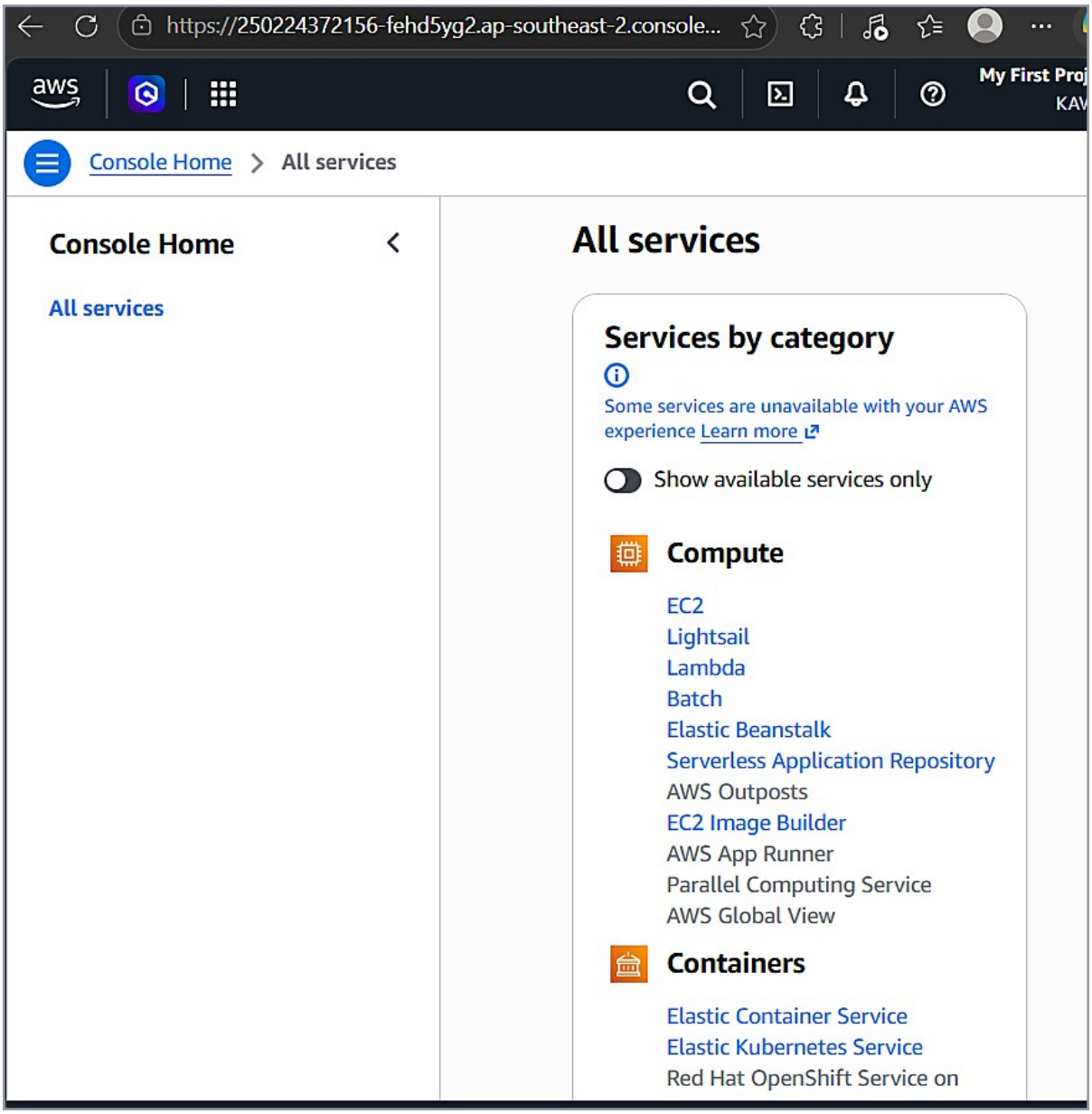
3. AWS Account and Region

All resources for this assignment were created from the AWS Management Console in the Asia Pacific (Sydney) region, ap-southeast-2.

AWS Console Home



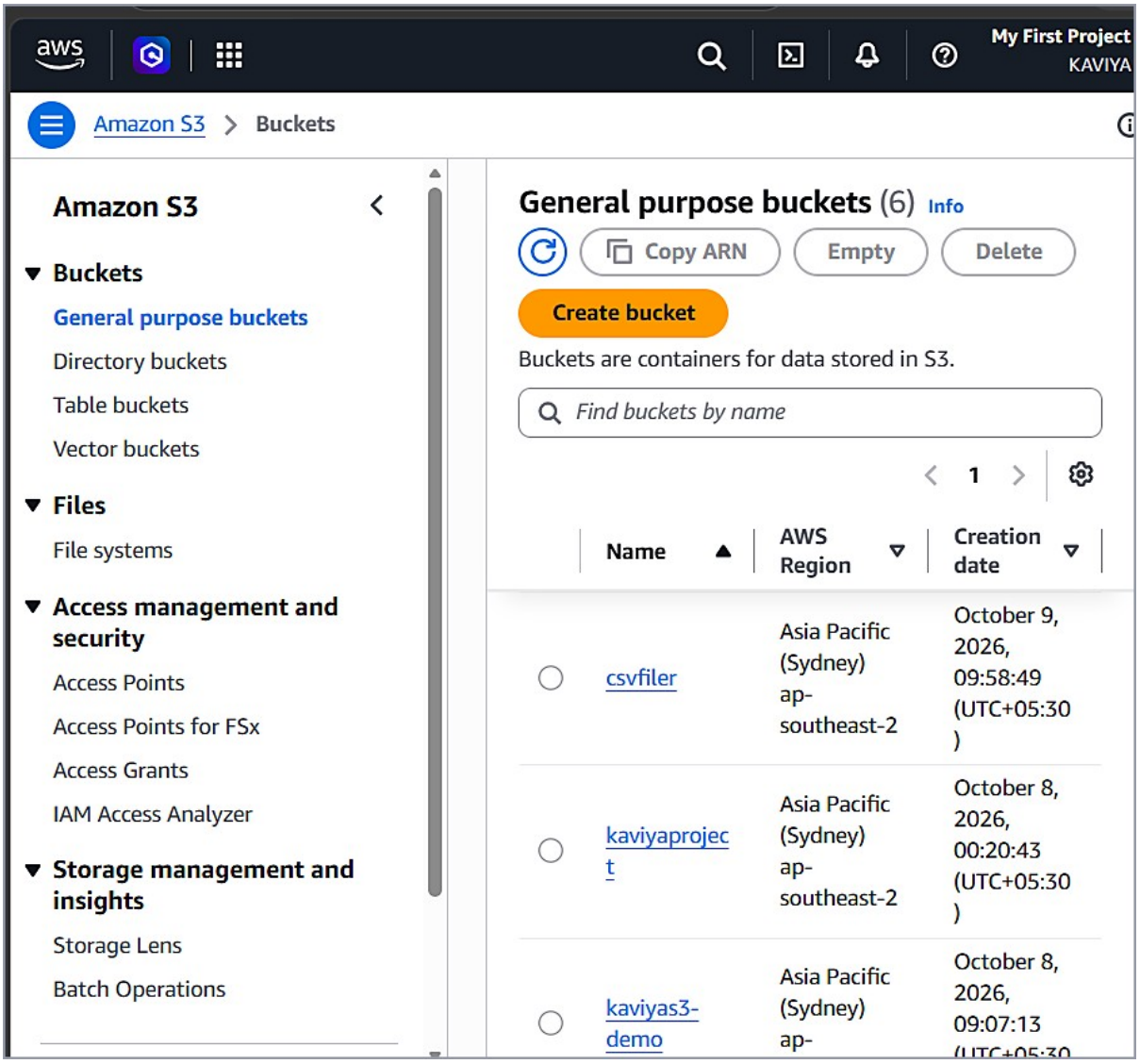
Selected AWS region (ap-southeast-2 shown in the console URL)



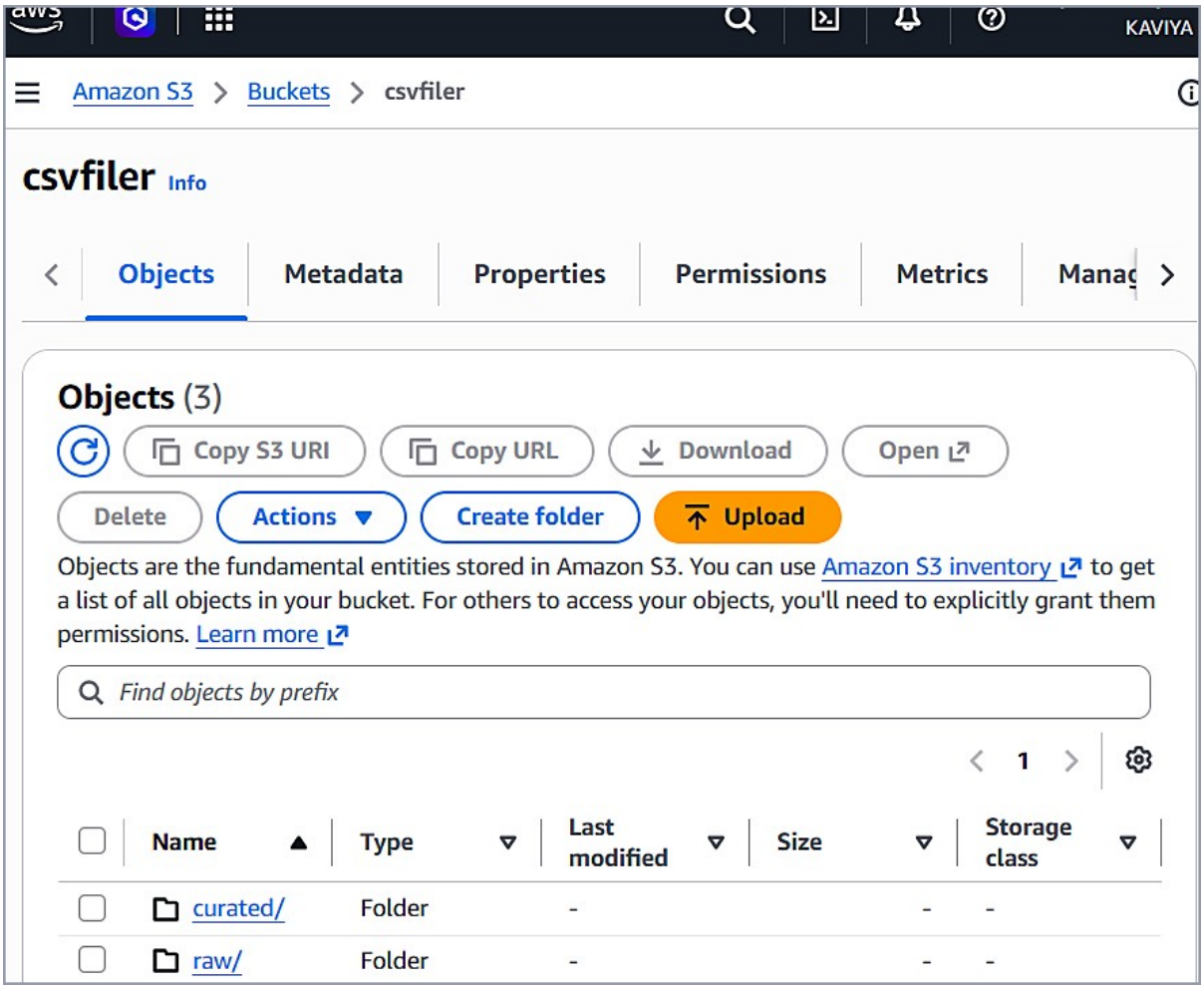
4. Amazon S3

An S3 bucket named csvfiler was created with two folders: raw/ for the original CSV data and curated/ for the transformed output.

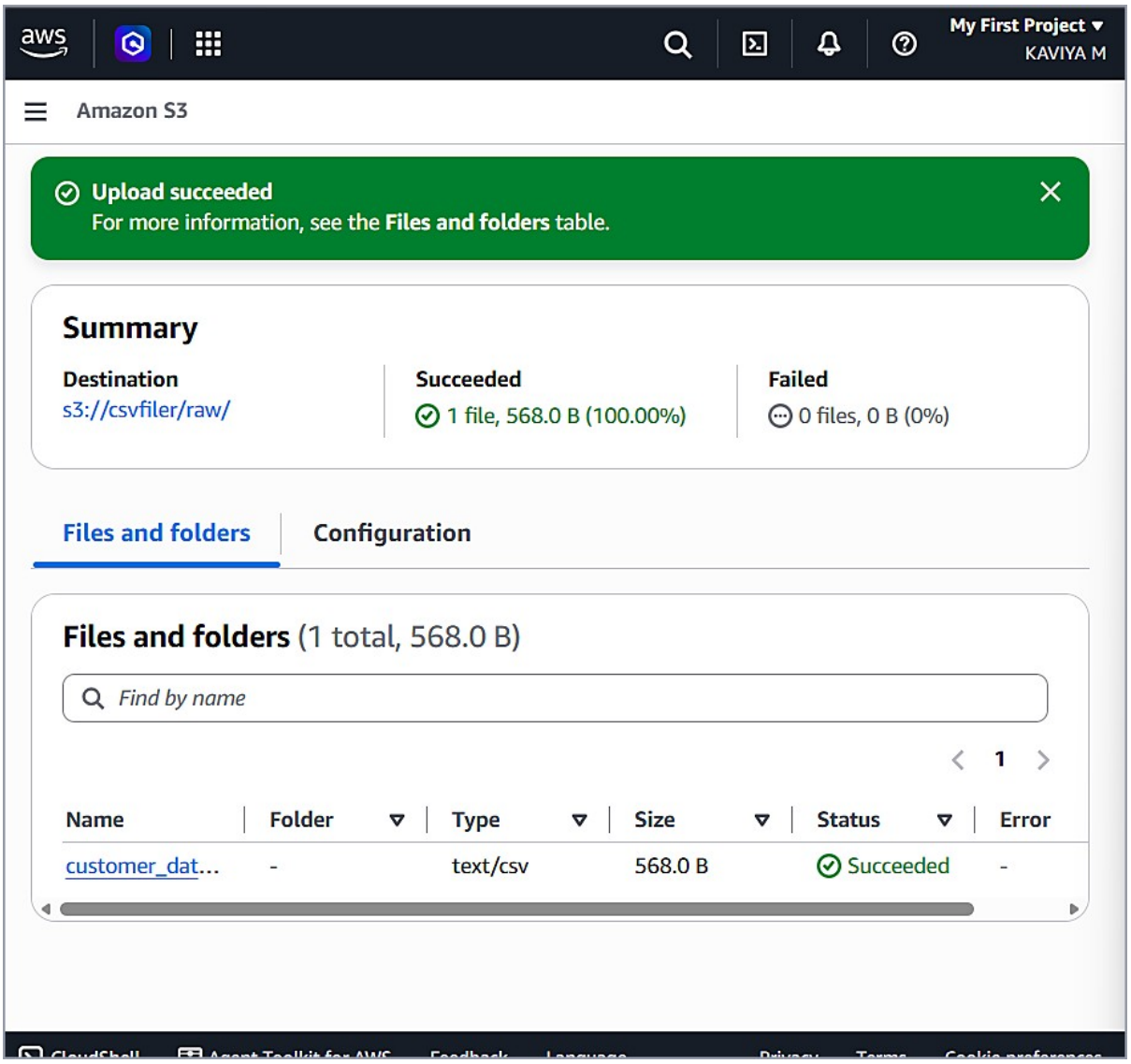
S3 bucket created: csvfiler (Asia Pacific, Sydney)



raw/ and curated/ folders inside the csvfiler bucket



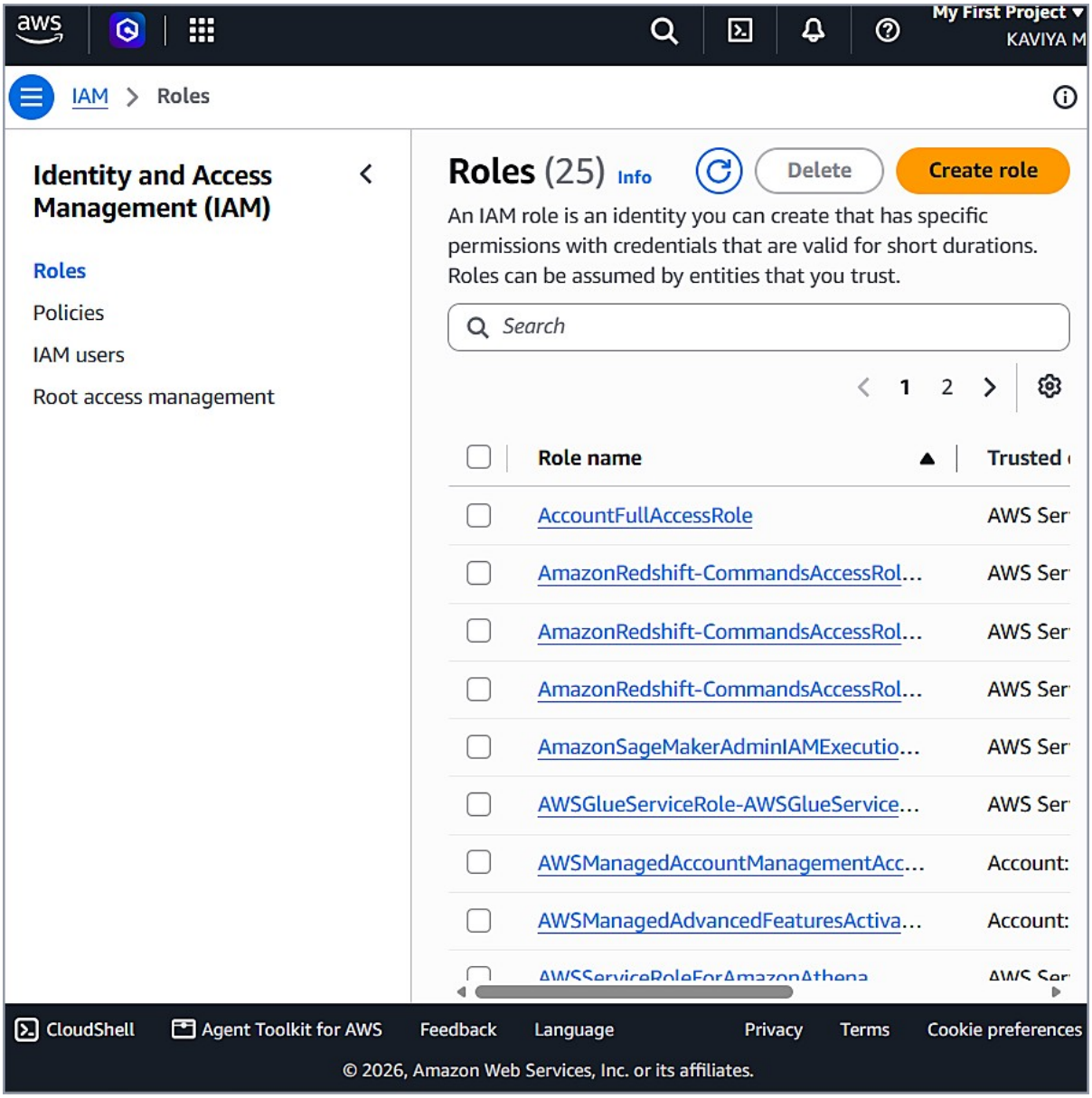
Upload to s3://csvfiler/raw/ completed successfully



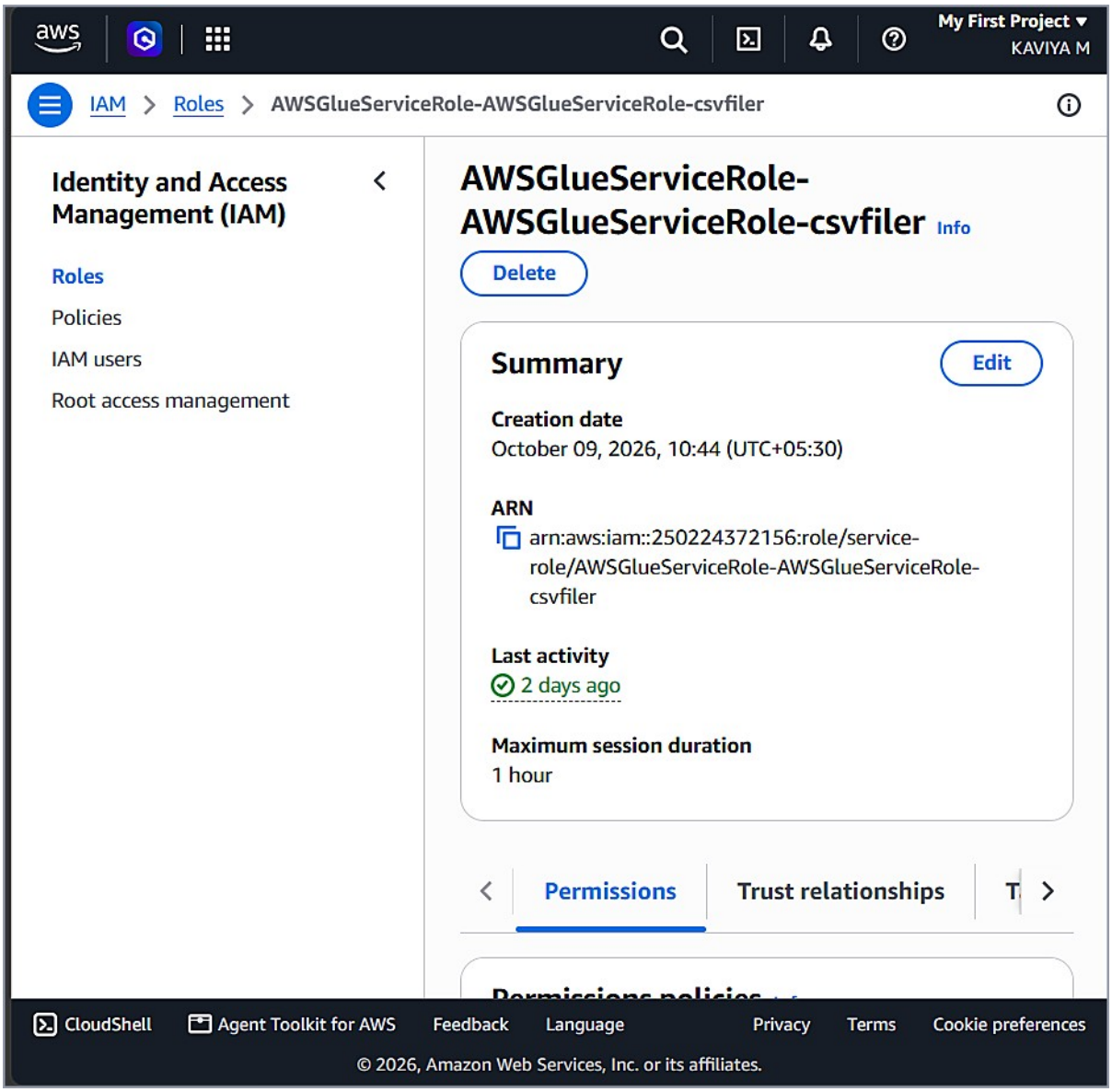
5. IAM Role for AWS Glue

The AWS Glue service role AWSGlueServiceRole-AWSGlueServiceRole-csvfiler was used by the crawler and the ETL job to read from and write to S3.

IAM roles list

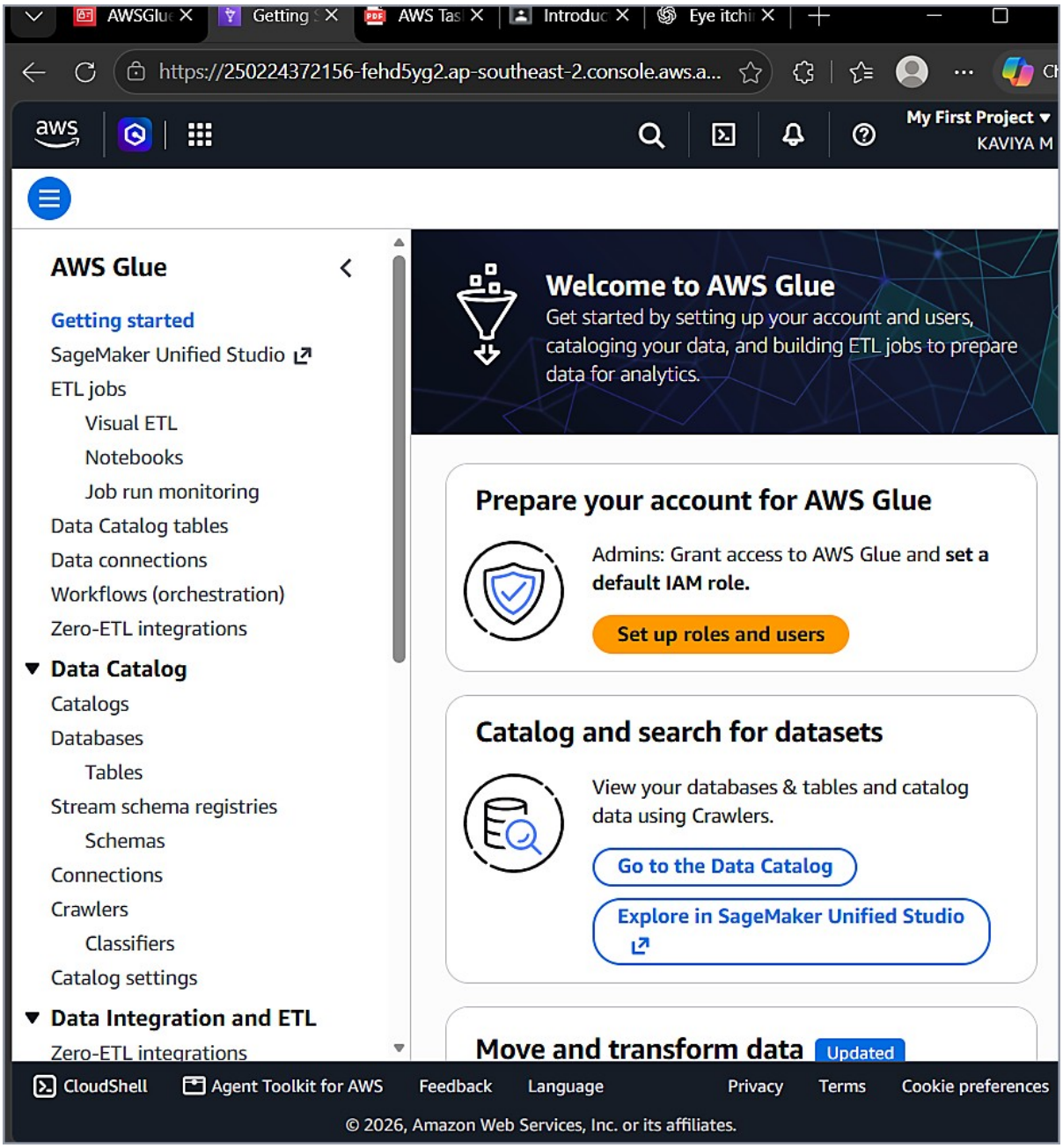


IAM role: AWSGlueServiceRole-AWSGlueServiceRole-csvfiler



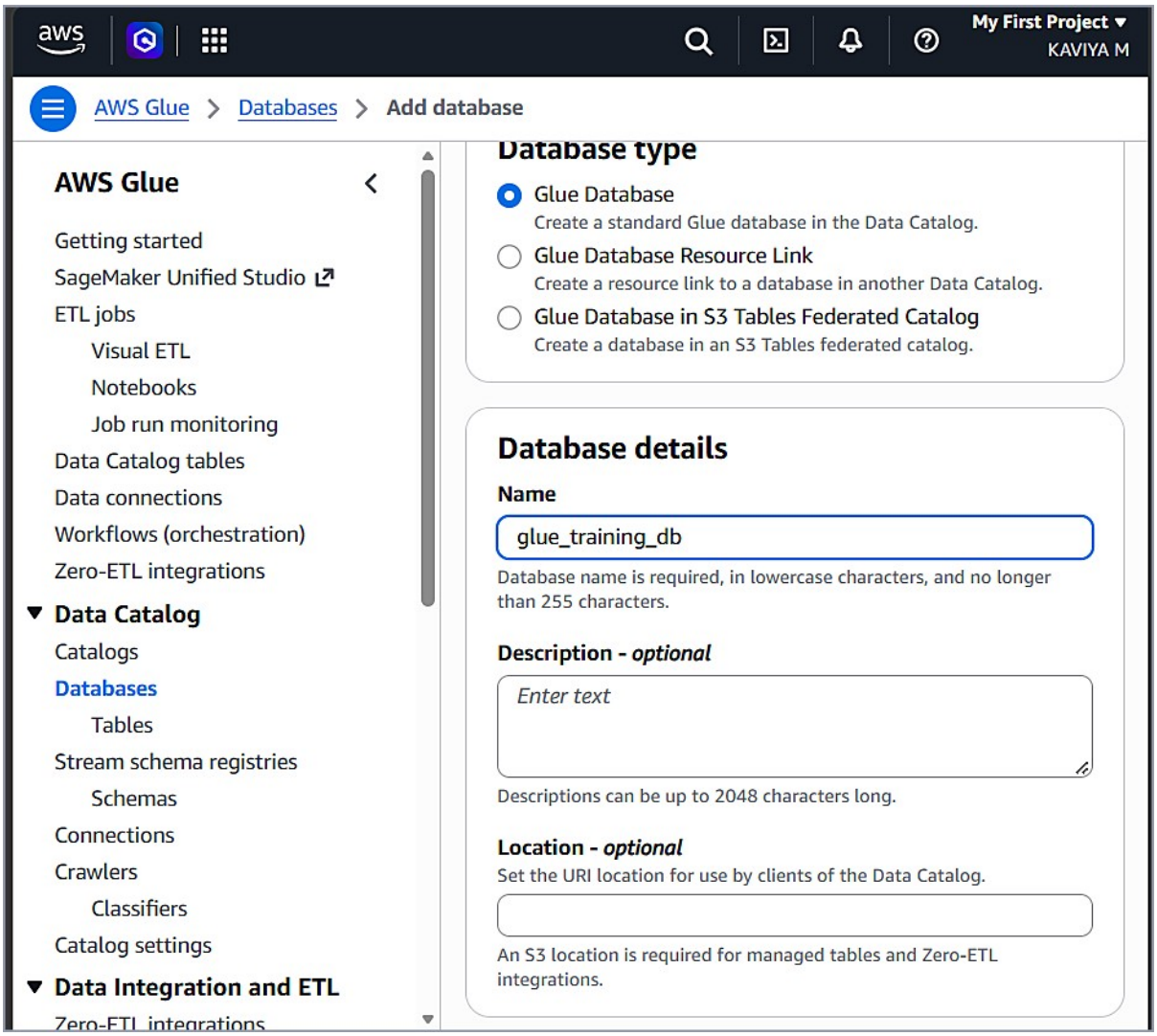
6. AWS Glue Console and Database

AWS Glue console

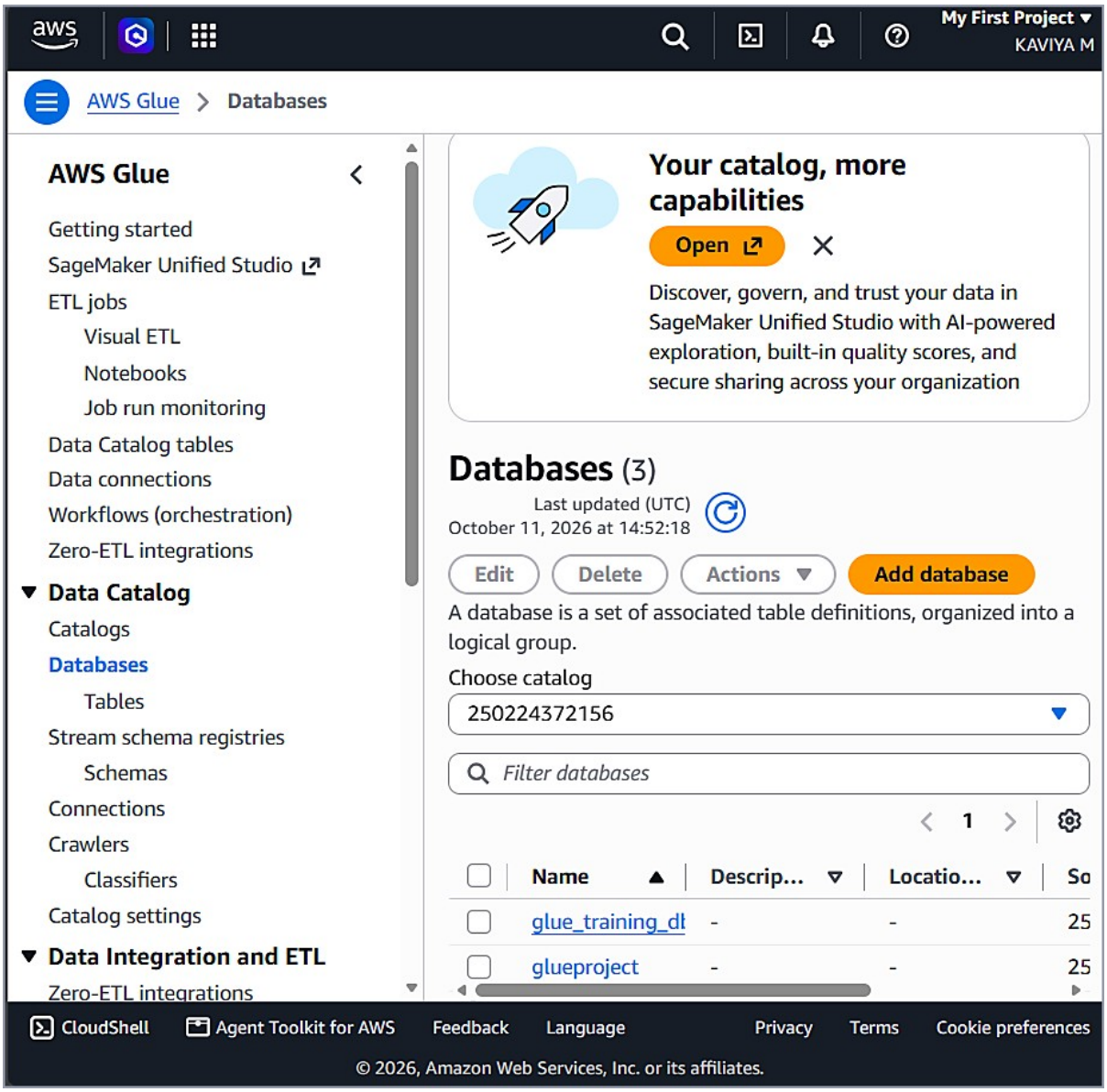


A database named glue_training_db was created in the AWS Glue Data Catalog.

Creating the database glue_training_db



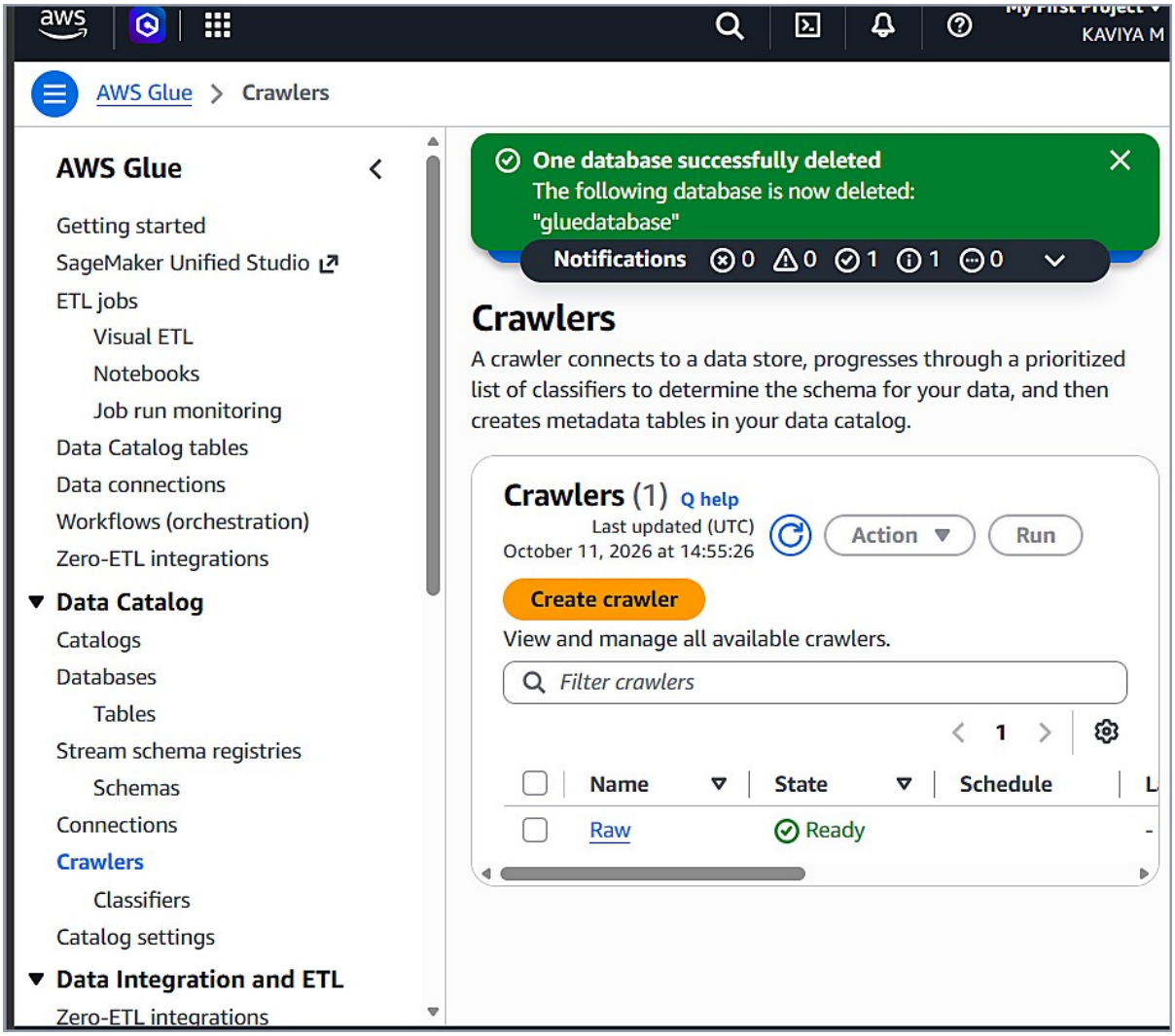
glue_training_db listed under Data Catalog databases



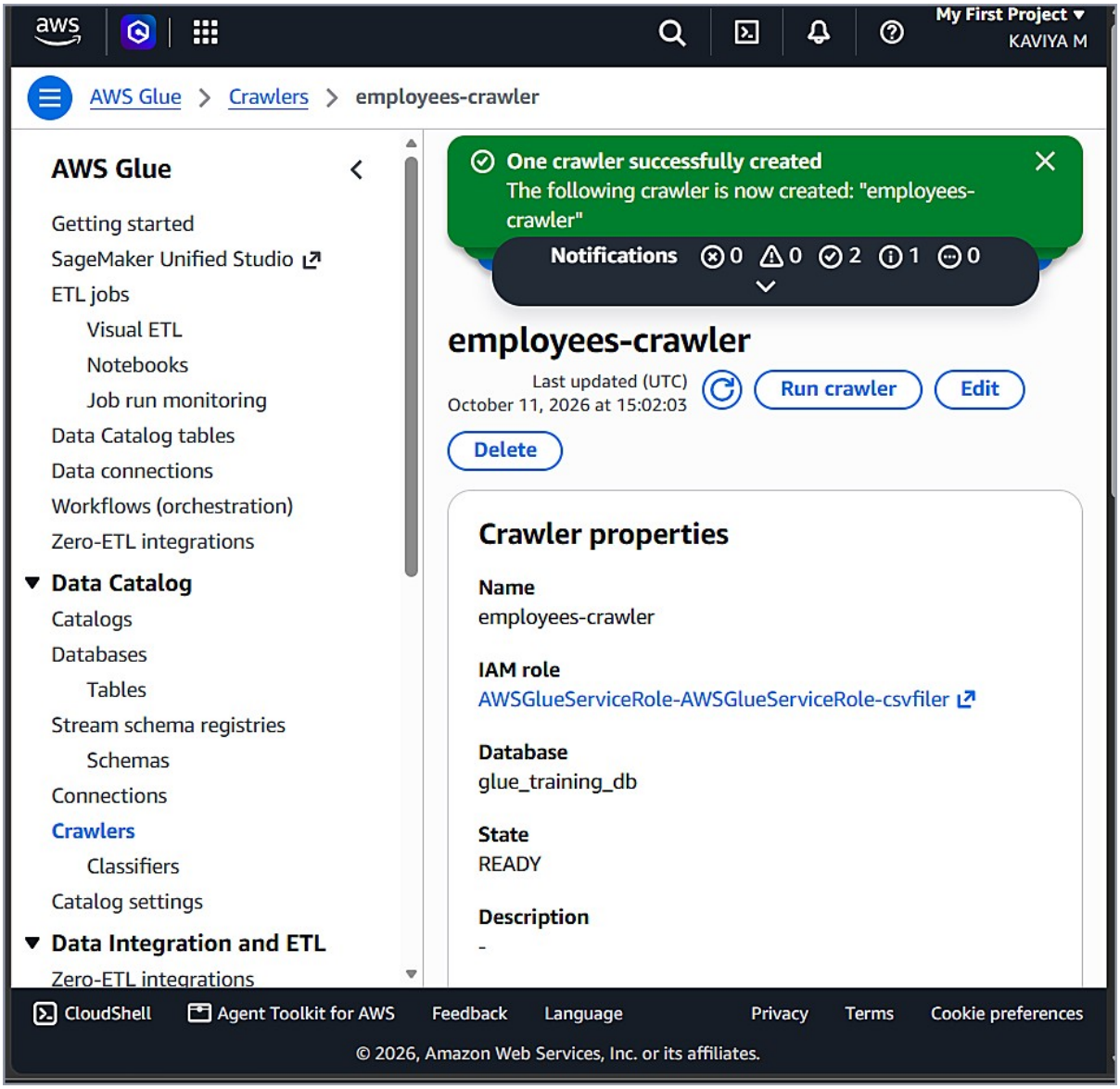
7. AWS Glue Crawler

A crawler named employees-crawler was created to scan the raw/ folder of the csvfiler bucket, using the Glue service role and storing its output in glue_training_db.

Crawlers page

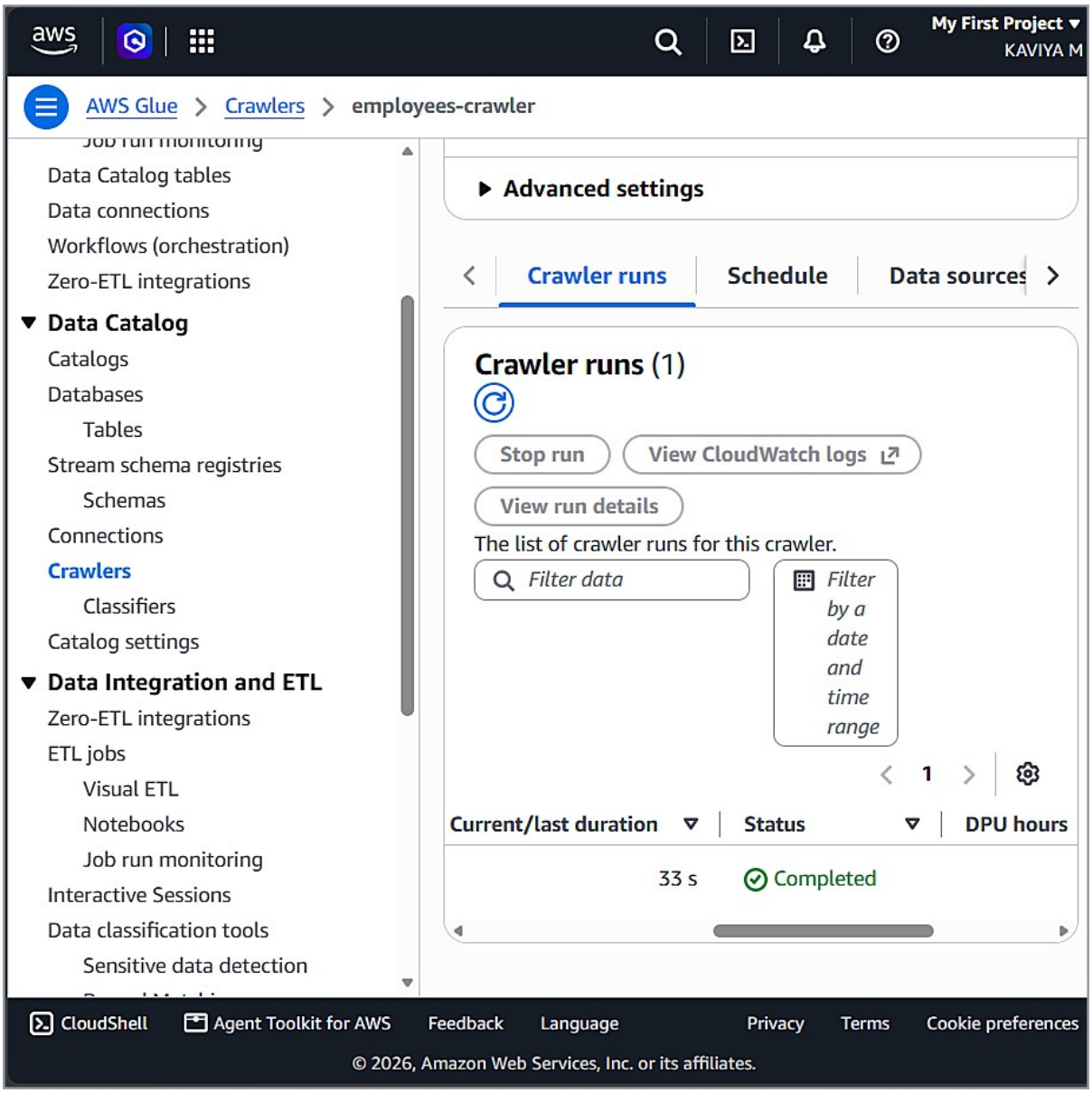


Crawler properties: name, IAM role, and target database



8. Crawler Execution

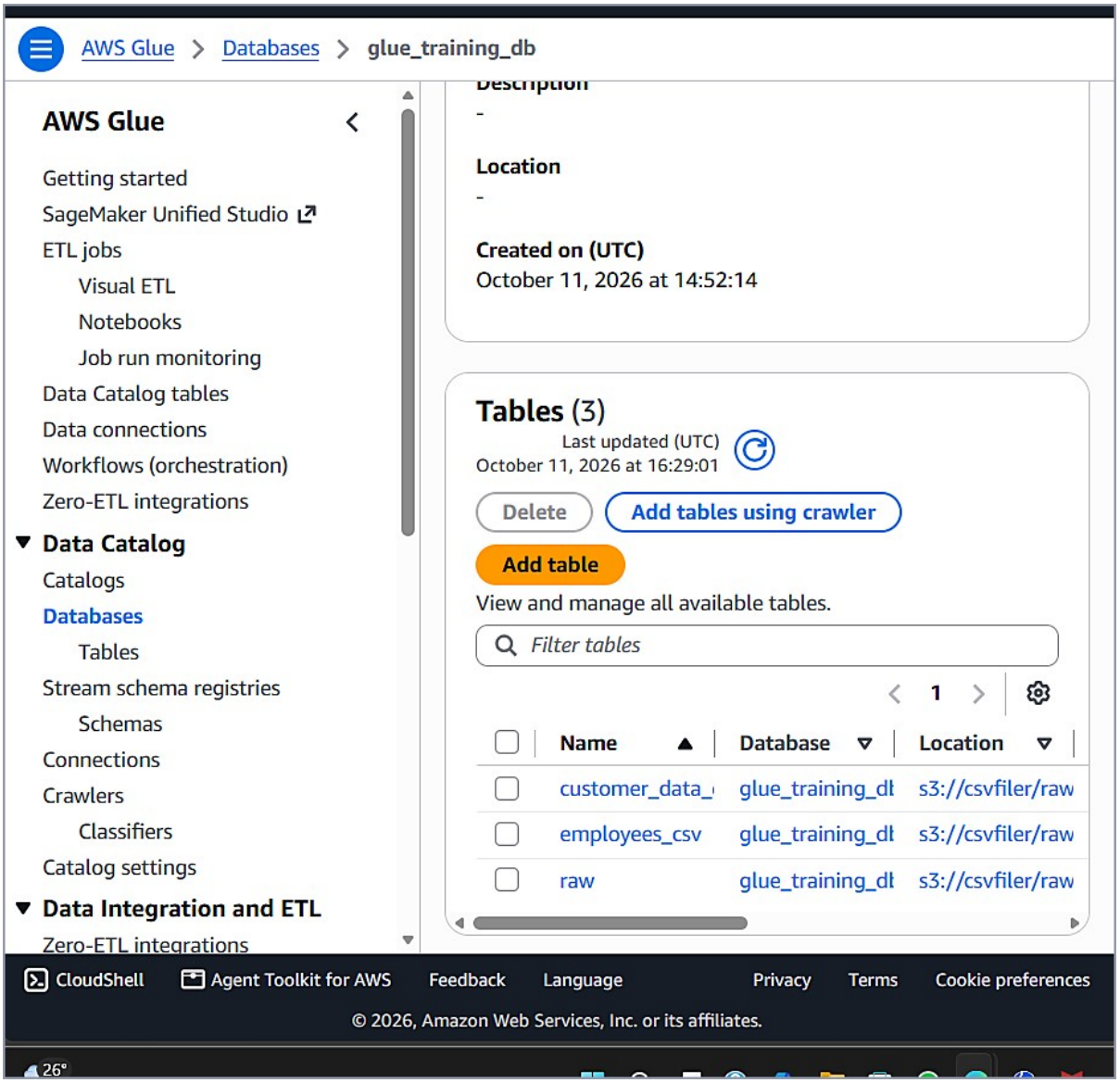
Crawler run status: Completed



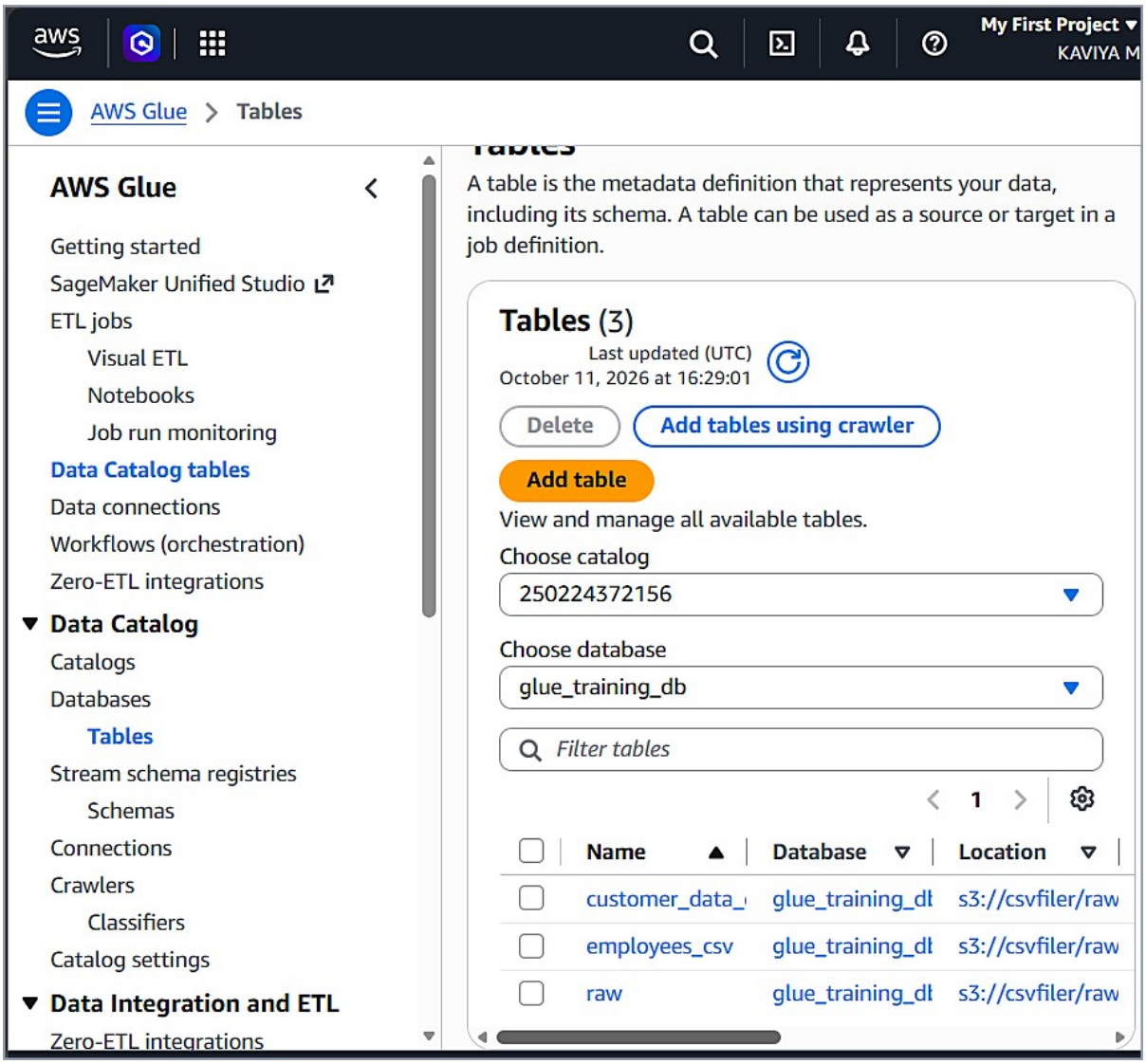
9. Glue Data Catalog

After the crawler ran, the table metadata was created in the glue_training_db database, including the employees_csv table.

Database glue_training_db with its tables



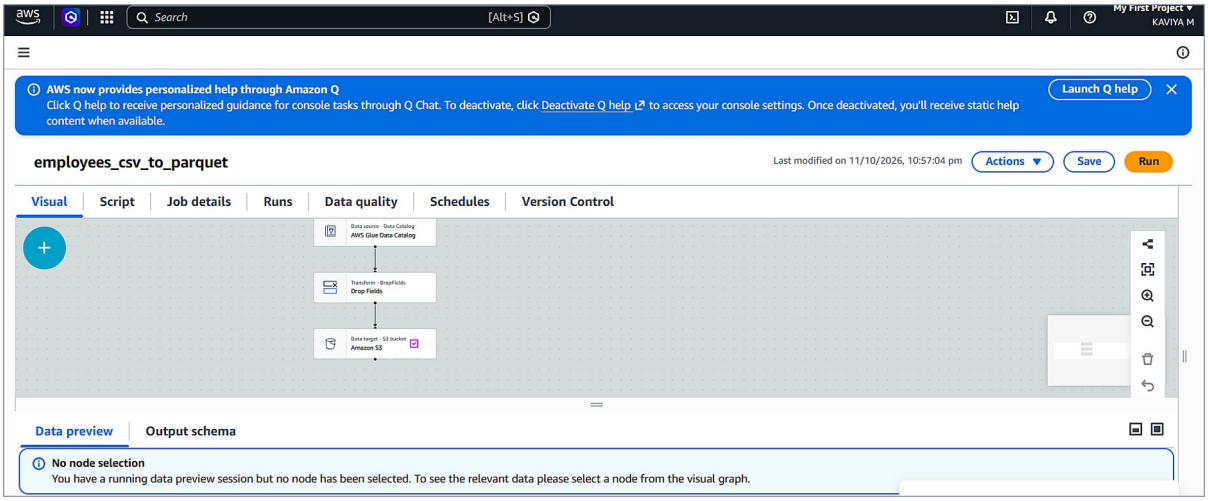
Tables successfully created in the Data Catalog



10. Glue Visual ETL

The job employees_csv_to_parquet was built in Glue Visual ETL. The complete workflow is: Data Catalog (employees_csv) → Drop Fields → Amazon S3 target.

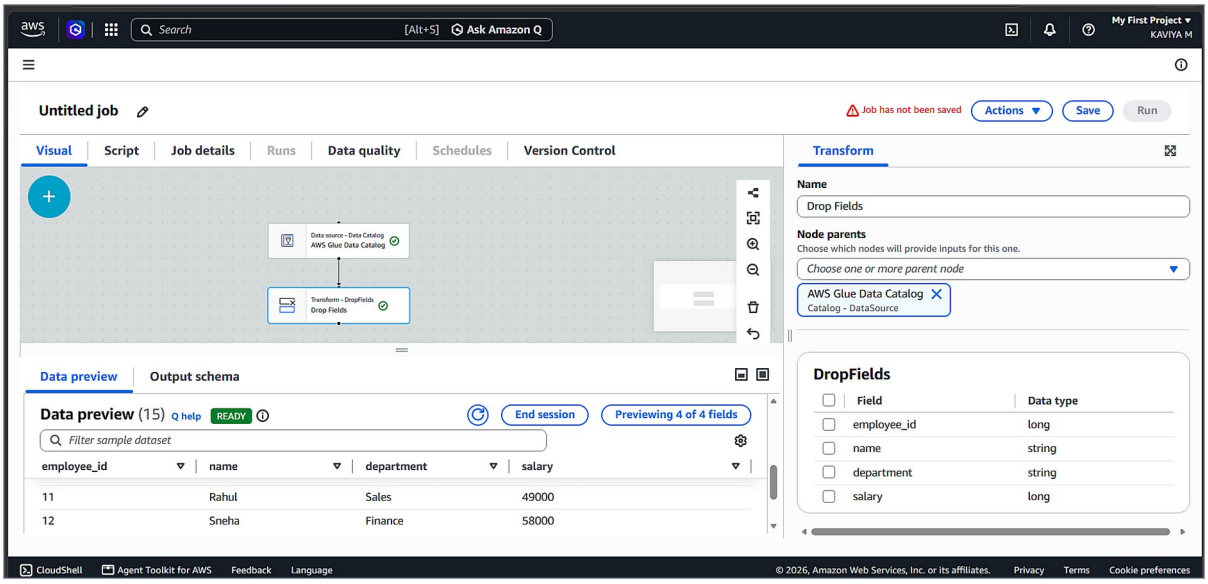
Visual ETL canvas: Data Catalog → Drop Fields → S3 target



11. Glue Transformation

A Drop Fields transformation removes the city column. The remaining columns are employee_id, name, department, and salary.

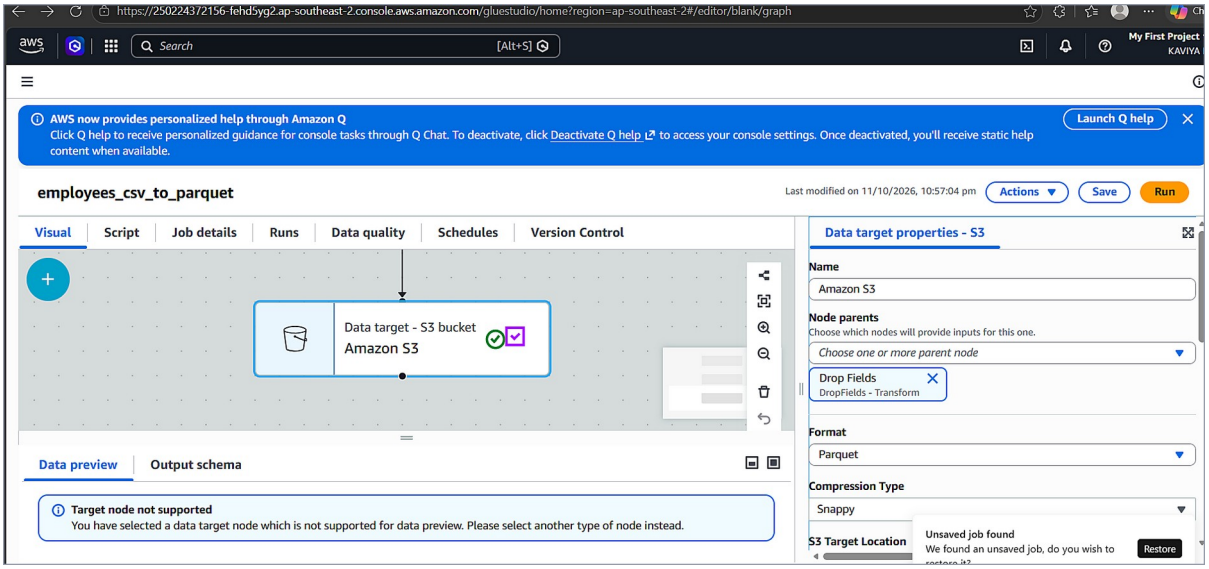
Drop Fields transformation configuration and data preview



12. S3 Target Configuration

The transformed data is written to the curated/ location of the csvfiler bucket in Parquet format with Snappy compression.

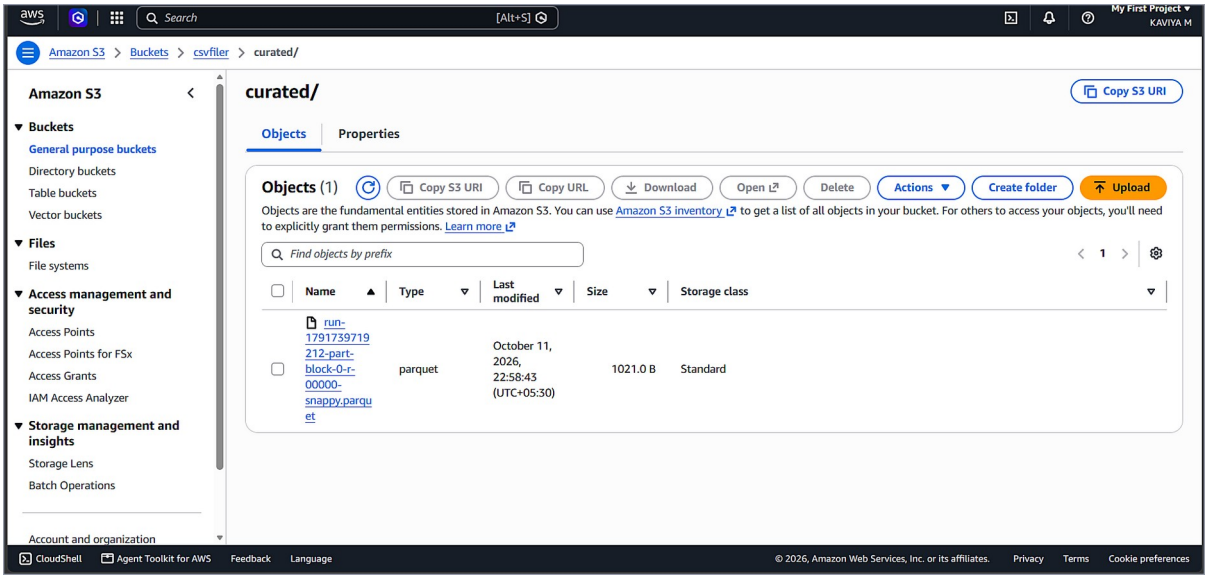
Data target properties: format Parquet, compression Snappy



13. Final S3 Output

The Parquet output file generated by the ETL job is stored in the curated/ folder.

csvfiler/curated/ containing the generated Parquet file



14. Final Explanation: AWS Cloud & Data Engineering

1. Amazon S3 – Raw Data Storage

Amazon S3 is used to store the employee dataset in CSV format. The file employees.csv is uploaded to the raw/ folder inside the csvfiler bucket. This folder contains the original data before any transformations are performed.

2. AWS Glue Crawler

The AWS Glue Crawler scans the raw data folder in Amazon S3. It automatically detects the CSV file format, identifies the column names, and determines the data types. The crawler then creates or updates table metadata based on the structure of the source data.

3. AWS Glue Data Catalog

The AWS Glue Data Catalog stores metadata about the employee dataset in the glue_training_db database. The table contains five columns: employee_id, name, department, salary, and city. This metadata helps AWS Glue understand how to read and process the source data.

4. AWS Glue Visual ETL

AWS Glue Studio Visual ETL provides a graphical interface to create data processing workflows without writing extensive code. The source is selected from the Glue Data Catalog table, allowing the job to read the employee records stored in Amazon S3.

5. Data Transformation

The DropFields transformation is used to remove the city column from the dataset. The remaining columns are employee_id, name, department, and salary. This step prepares the data according to the requirements of the assignment.

6. Amazon S3 – Curated Data Storage

After transformation, the processed data is directed to the curated/ folder in the csvfiler S3 bucket. Separating raw and curated data helps organize the original dataset and the processed output.

7. Parquet Output Format

The output format is configured as Parquet. Parquet is a column-oriented file format that supports efficient storage and analytical queries. It is commonly used in data engineering workflows because it can reduce storage requirements and improve query performance.

8. Execution and Final Result

The Glue ETL job processes the employee data according to the configured workflow. When the job runs successfully, the transformed dataset is written to the curated S3 location in Parquet format. The resulting data contains only the four required columns and can be used for further analysis and reporting.

Conclusion

This workflow demonstrates how Amazon S3 and AWS Glue can be used together to build a simple cloud-based data engineering pipeline. The process moves data from raw storage through metadata discovery and transformation into curated storage. It helps make the dataset more organized and ready for analytical use.